

RIVER WATER QUALITY IN HONG KONG IN 2017



Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Mission

To conduct an effective and scientifically sound water monitoring programme that helps protect the health of Hong Kong's rivers and streams and ensure the water quality objectives can be achieved and maintained.



Tin Shui Wai Nullah

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1. Introduction

Rivers and streams in Hong Kong are generally short in length and carry small flows. Many of them have their upstream sections located within water gathering grounds and have been drawn on quite extensively for supplies of potable water. At present, other than those located within water gathering grounds, the primary beneficial uses of the rivers and streams in Hong Kong include the maintenance of aquatic life, general amenity, flood prevention and storm water drainage. The main channel of Shing Mun River is the only watercourse currently used for secondary contact recreation.

The Environmental Protection Department (EPD) has initiated a comprehensive river water quality monitoring programme since 1986. It mainly monitors the major rivers flowing through the urban areas, with one or more representative monitoring stations set up at the upstream and downstream sections of the main channels and tributaries. A number of small rivers and streams in the New Territories are also monitored for water pollution control and management purposes.

The monitoring programme serves the following purposes:

- evaluate the pollution status of river waters;
- monitor long-term changes in water quality;
- provide a scientific basis for planning water pollution control strategies;
- assess the compliance with the key statutory Water Quality Objectives (WQOs); and
- compile Water Quality Index (WQI).

This report summarises the water quality of rivers and streams covered by EPD's river monitoring programme in 2017. Annual reports can be downloaded from the EPD's website:

<http://wqrc.epd.gov.hk/en/water-quality/river-2.aspx>



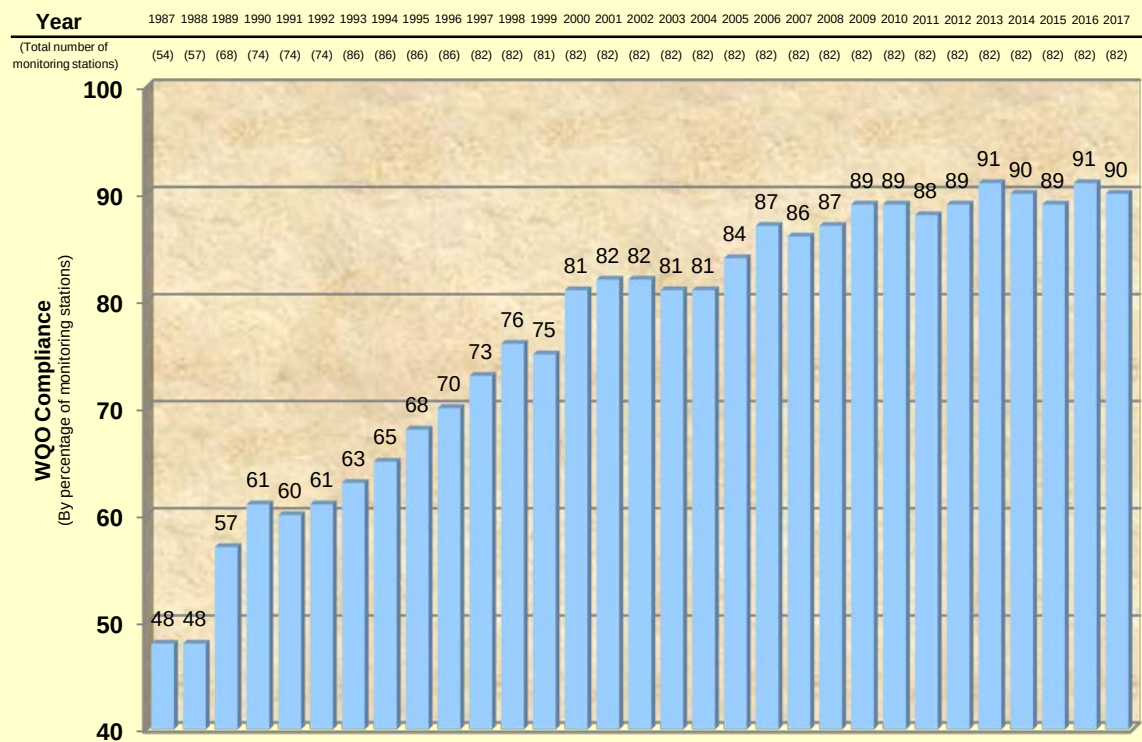
Sample collection at Tai Wai Nullah

2. An overview of Hong Kong's rivers in 2017

In 2017, the EPD's river monitoring programme covered 30 watercourses and a total of 82 monitoring stations (Appendix A), as compared with some 14 watercourses and 47 stations when the programme started in 1986. The monitoring programme involved monthly conducting field measurements and collecting water samples for laboratory analyses. Over 40 physico-chemical and biological parameters, including organic matters, nutrients, metals and *E. coli* were analysed (Appendix B).

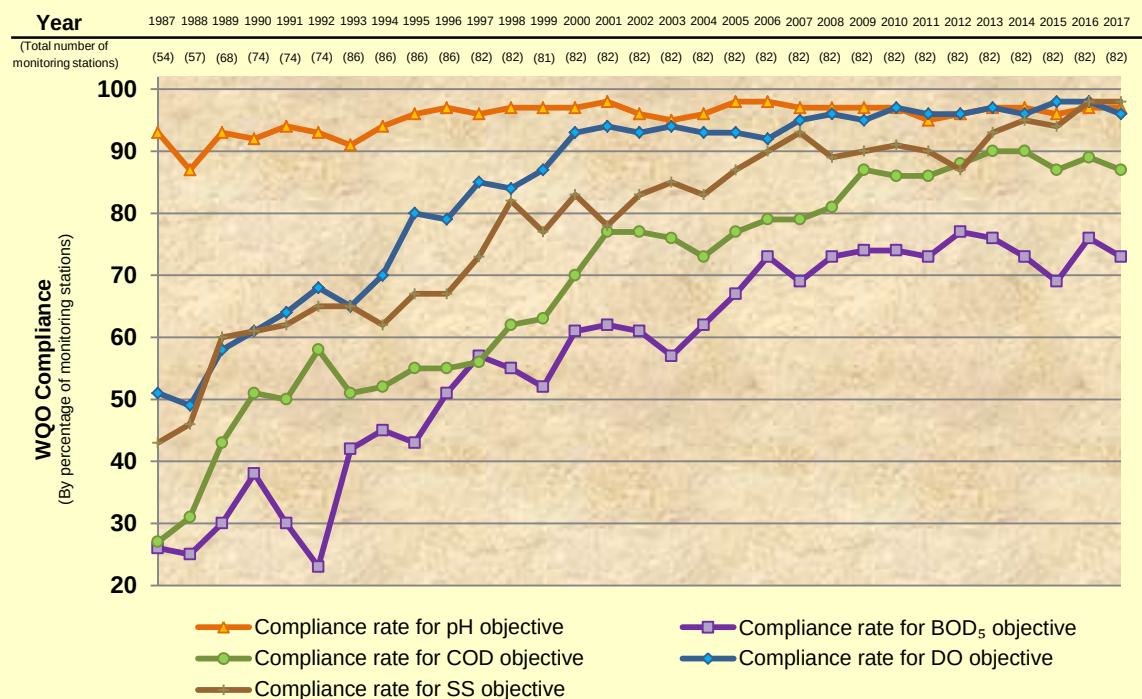
To assess compliance with the statutory Water Quality Objectives (WQOs), five representative parameters are used, namely pH, suspended solids, dissolved oxygen, 5-day biochemical oxygen demand and chemical oxygen demand (Appendix C). This report presents the annual average compliance rates for individual watercourses as well as overall compliance rates for Hong Kong (Appendix F). For rivers that have more than one monitoring stations, the WQO compliance was expressed as an average of the annual compliance rates of all their individual monitoring stations where full compliance would be indicated by 100%. Similarly, the overall compliance rates for the entire Hong Kong were the averages of that of all the monitoring stations in the territory. The water quality of inland waters in Hong Kong in 2017 continued to be good with an overall compliance rate at 90%, compared with 91% in 2016 and 89% in 2015. The high compliance rates in recent years were the result of the implementation of pollution control legislation, including the Water Pollution Control Ordinance and the Livestock Waste Control Scheme introduced under the Waste Disposal Ordinance, as well as the extension of the sewer network to more villages under the Sewage Master Plans.

Overall WQO compliance in the inland waters of Hong Kong, 1987-2017



Figures are rounded to the nearest integer

WQO compliance for five representative parameters in the inland waters of Hong Kong, 1987-2017

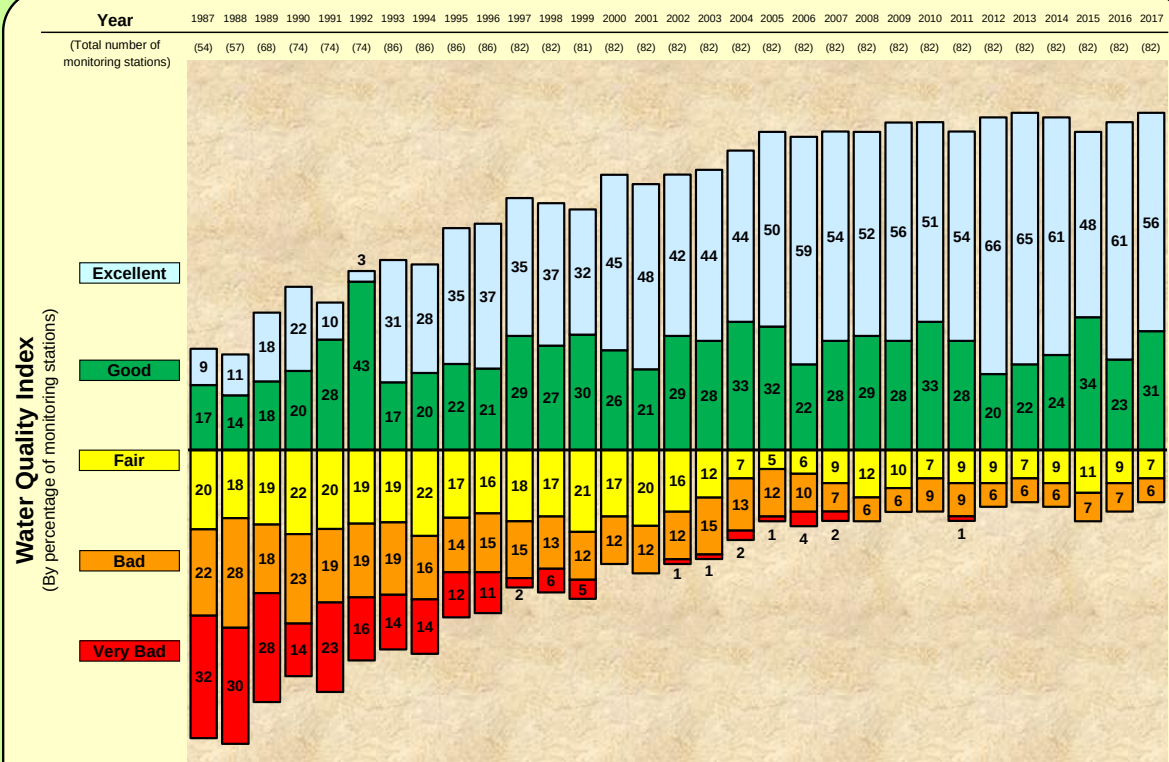


The Water Quality Index (WQI) is relevant to conserving the primary beneficial use of rivers and streams for the maintenance of aquatic life. It grades the general ecological health of rivers and streams into five categories of “Excellent”, “Good”, “Fair”, “Bad” and “Very Bad” by assessing three key water quality parameters: dissolved oxygen, 5-day biochemical oxygen demand and ammonia-nitrogen content¹. In 2017, 87% of the river monitoring stations were graded “Good” or “Excellent”, as compared with 26% in 1987, suggesting that the river water quality has greatly improved and the pollution loadings in these watercourses have been significantly lowered over the past three decades. The majority of the monitoring stations in Lantau Island, the Eastern New Territories, Southwestern New Territories and Kowloon were in these two categories. On the other hand, only 6% of the monitoring stations were graded “Bad” and no station was graded “Very Bad” in 2017, as compared with 22% “Bad” and 32% “Very Bad” in 1987. Most of the “Bad” stations are located along watercourses in the Northern and Western New Territories. Compared with 2016, six stations moved up one grade and seven stations moved down one grade in 2017. These minor changes are considered to be within the normal range of natural fluctuations over the past ten years.

River monitoring station	2016 WQI grading	2017 WQI grading
Kai Tak River (KN1, KN2)	Fair	Good
Kau Wa Keng Stream (KW3)	Good	Excellent
Tung Mun River (TN4, TN5)	Good	Excellent
Yuen Long Creek (YL2)	Bad	Fair
Mui Wo River (MW5)	Excellent	Good
River Ganges (GR2)	Excellent	Good
Shing Mun River (TR17, TR17L, TR19)	Excellent	Good
Tin Shui Wai Nullah (TSR2)	Excellent	Good
Tung Chung River (TC3)	Excellent	Good

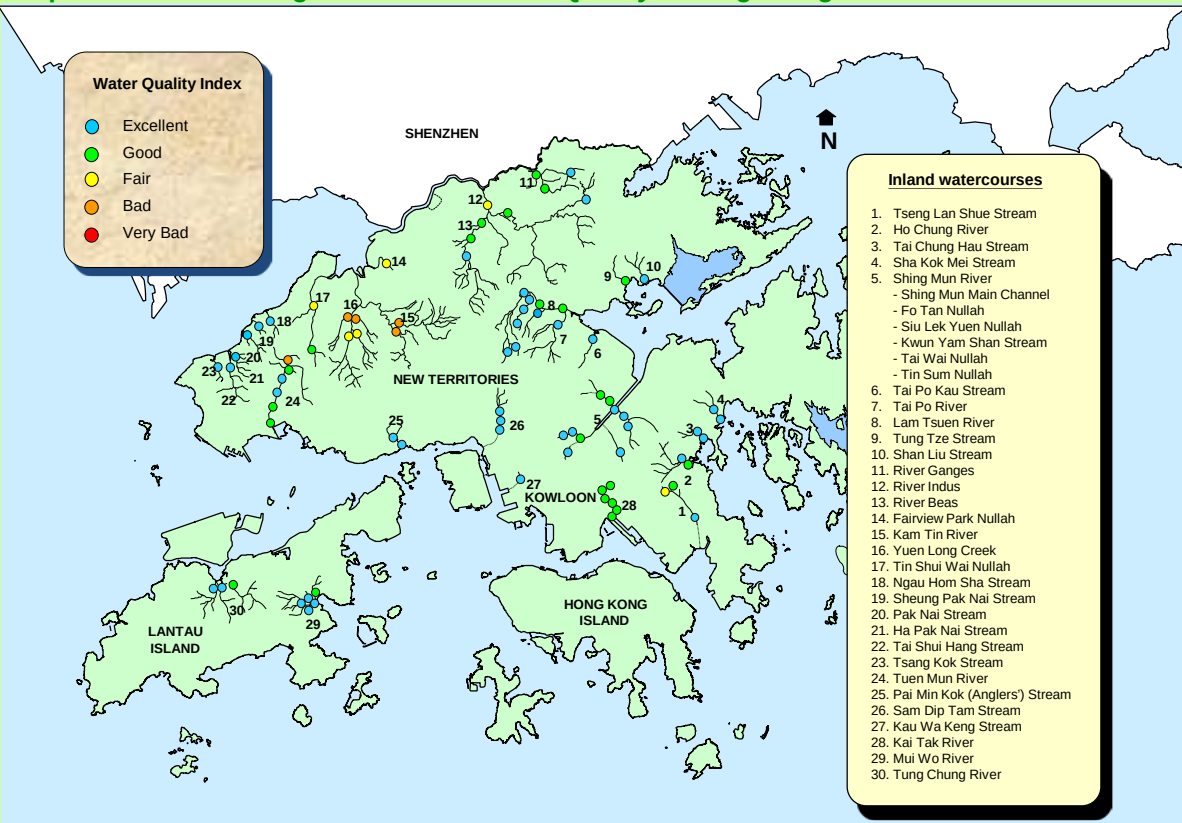
¹ Please refer to Appendix G for more details on the calculation and assessment of Water Quality Index.

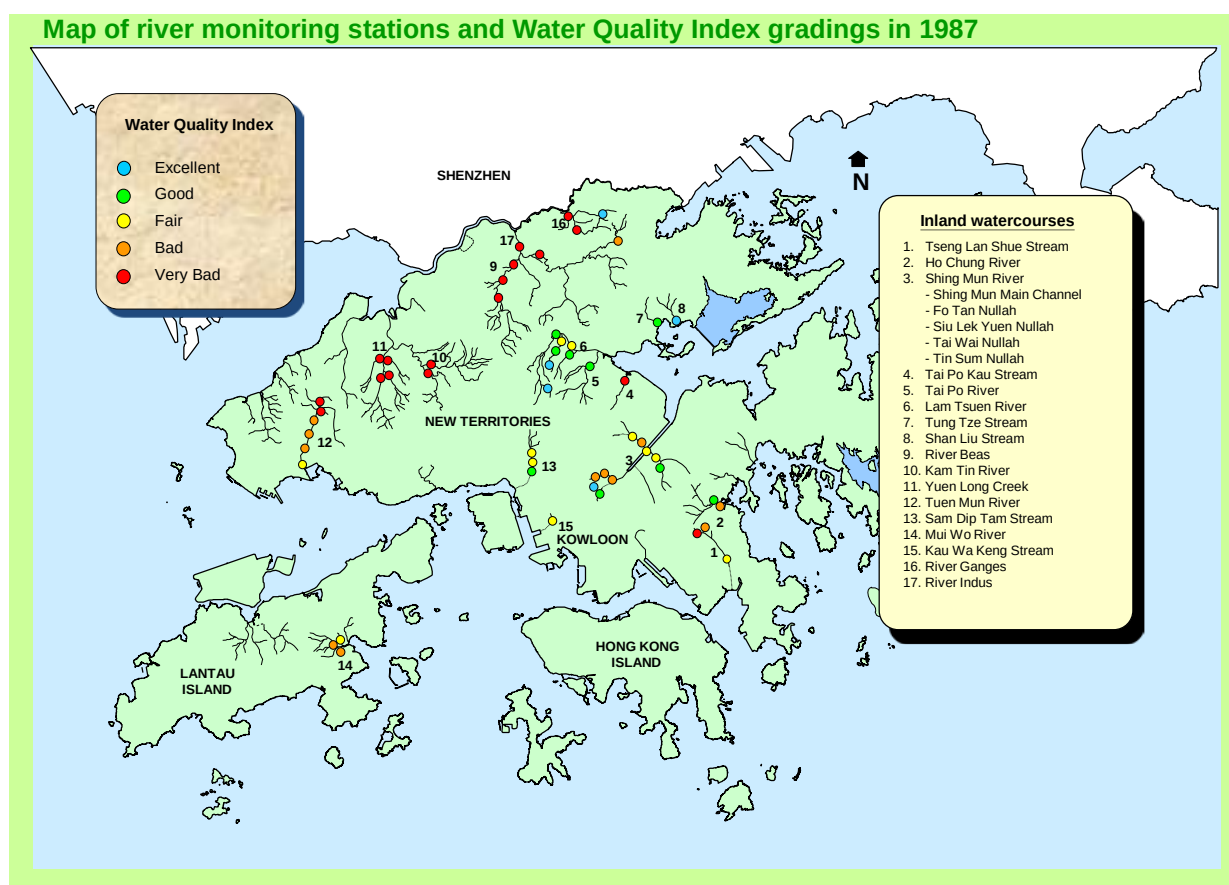
Water Quality Index gradings for the inland waters of Hong Kong, 1987-2017



Figures are rounded to the nearest integer

Map of river monitoring stations and Water Quality Index gradings in 2017





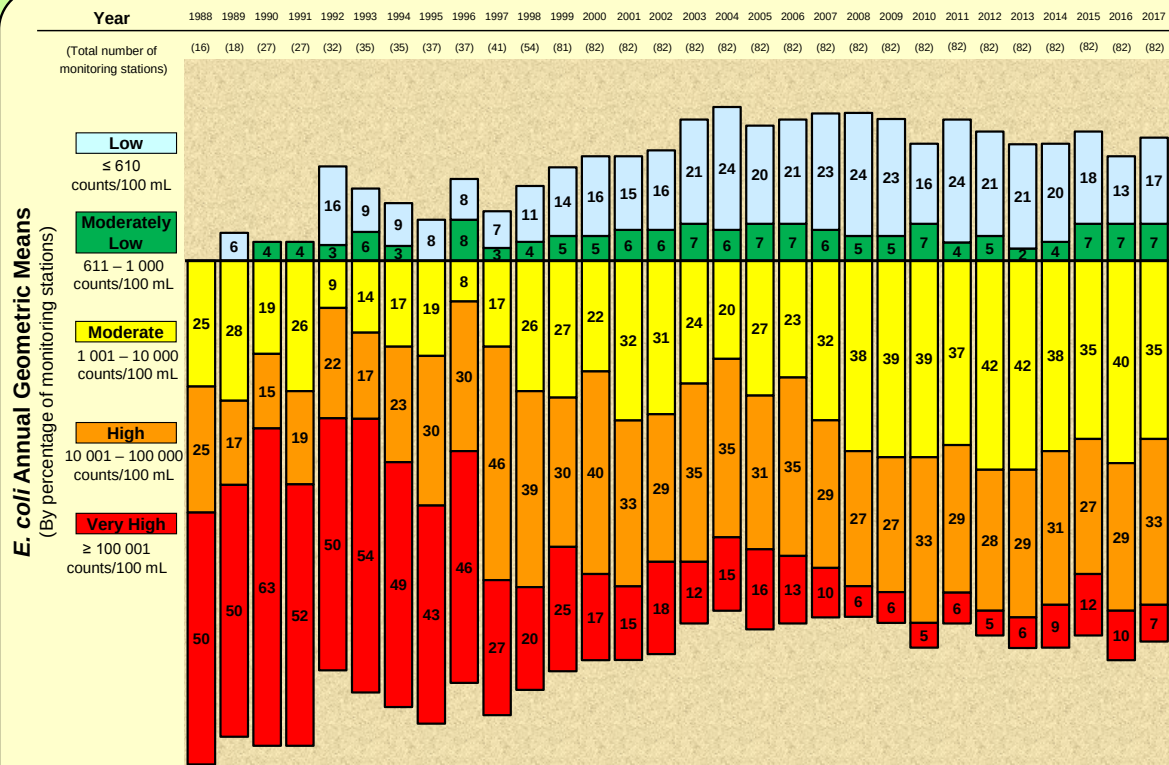
Since 1988, *E. coli* level in the river waters was also analysed. As faeces of all warm-blooded animals contain *E. coli*, its measured level in a water body is a common indicator of the presence and extent of faecal contamination. In 2017, 24% of the monitoring stations had “Low” or “Moderately Low” levels of *E. coli* (i.e. less than or equal to 1 000 counts/100mL) while 40% recorded “High” or “Very High” (i.e. over 10 000 counts/100mL) levels².

Most of the stations with “Very High” levels of *E. coli* are located in the western part of the New Territories (e.g. Yuen Long Creek and Kam Tin River), mainly due to discharges from livestock farms, runoff from unsewered village houses as well as expedient connections in the old districts. While these watercourses are primarily intended for stormwater conveyance and flood prevention purpose, the Government will continue its efforts in enforcement of pollution control legislations and extension of the public sewer network to reduce the *E. coli* levels at these locations.

In summary, the state of water quality of Hong Kong’s inland watercourses remained largely steady in 2017 and extended a long-term improvement trend as a result of the gradual reduction of pollution loading to the rivers and streams.

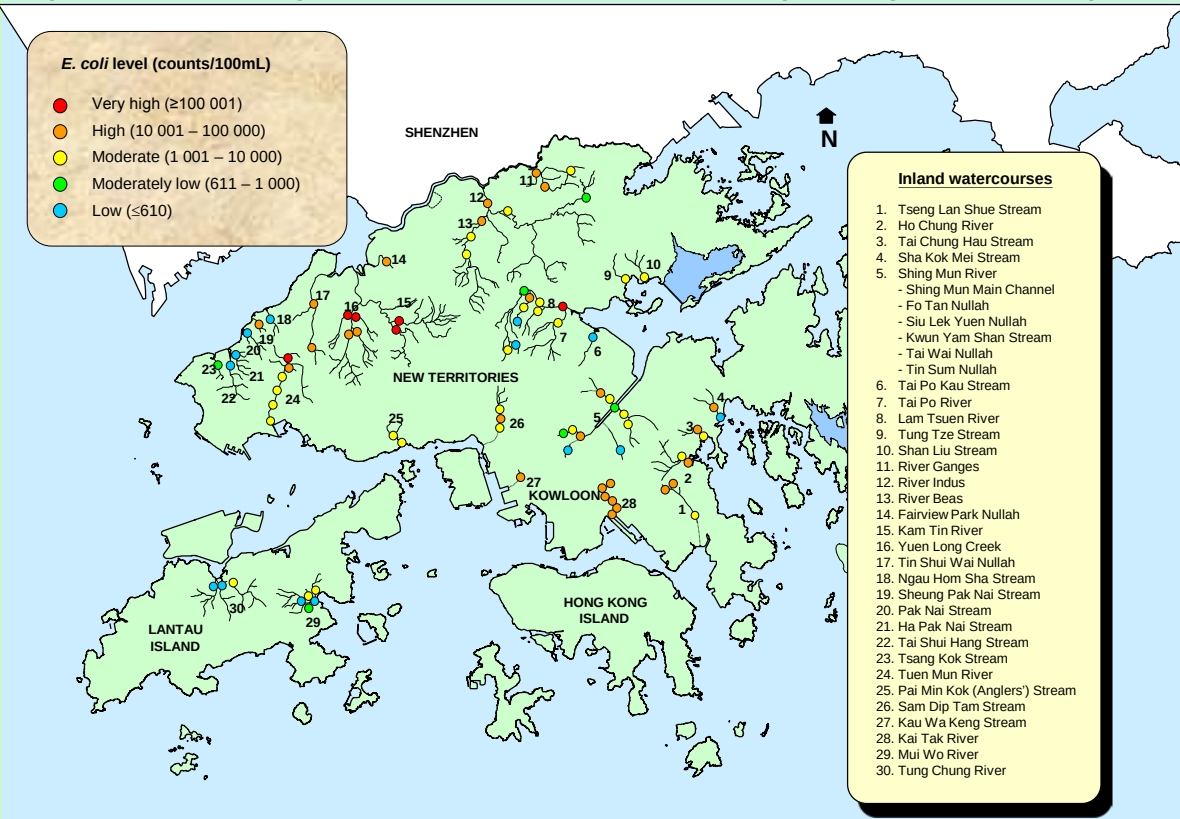
² All levels of *E. coli* in this report are reported as annual geometric means (counts/100mL).

Annual geometric mean *E. coli* levels for inland waters of Hong Kong (1988-2017)



Figures are rounded to the nearest integer

Map of river monitoring stations and levels of *E. coli* in 2017 (Annual geometric mean)



3. Eastern New Territories rivers and streams

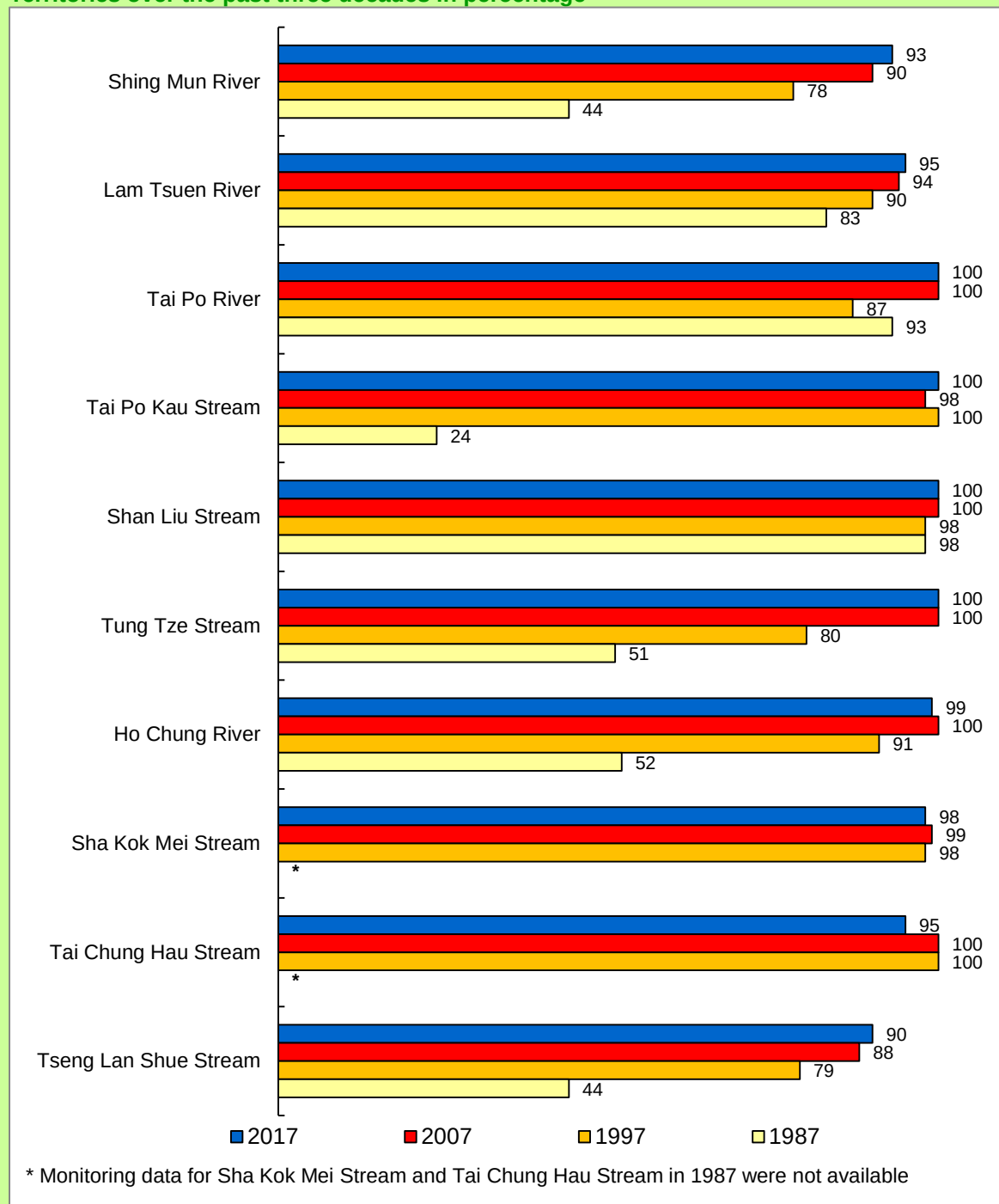
The EPD monitored ten watercourses in the Eastern New Territories in 2017. Six are located in the Tolo Harbour and Channel Water Control Zone (WCZ): Shing Mun River, Lam Tsuen River, Tai Po River, Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream. Within the Port Shelter WCZ, there are Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream. Tseng Lan Shue Stream is located within the Junk Bay WCZ.

The water quality of rivers and streams in the Eastern New Territories is good. In 2017, the overall WQO compliance rate of all monitoring stations in the Eastern New Territories was 95%, as compared with 94% in 2007, 86% in 1997 and 59% in 1987. Within the region, four rivers achieved full compliance (100%) with the WQOs in 2017. These are Tai Po River, Tai Po Kau Stream and Shan Liu Stream and Tung Tze Stream in the Tolo Harbour and Channel WCZ.



Shing Mun River

Compliance with the Water Quality Objectives of the rivers and streams in the Eastern New Territories over the past three decades in percentage



Shing Mun River, a major river which has three main tributaries and runs through the densely populated Sha Tin urban area, showed marked improvement during the past three decades. The WQO compliance rate of Shing Mun River was 93% in 2017. Lam Tsuen River, as a major river which passes through the urban area of Tai Po and joins with Tai Po River before entering Tolo Harbour, recorded a WQO compliance rate of 95% in 2017.



Tseng Lan Shue Stream upstream

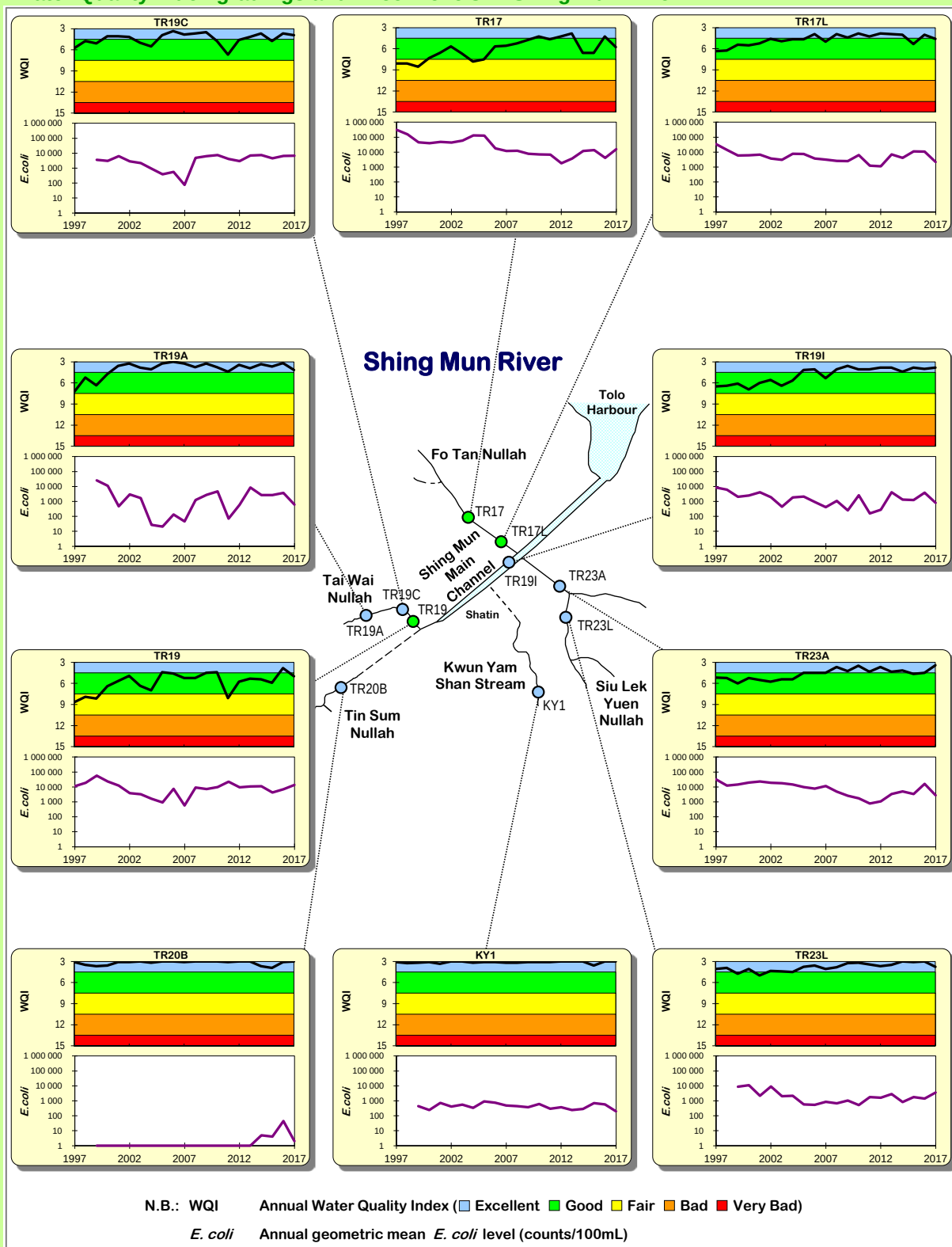
In Port Shelter WCZ, the WQO compliance for Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream were 99%, 98% and 95% respectively in 2017.

Tseng Lan Shue Stream in Junk Bay WCZ recorded a compliance rate of 90% in 2017.

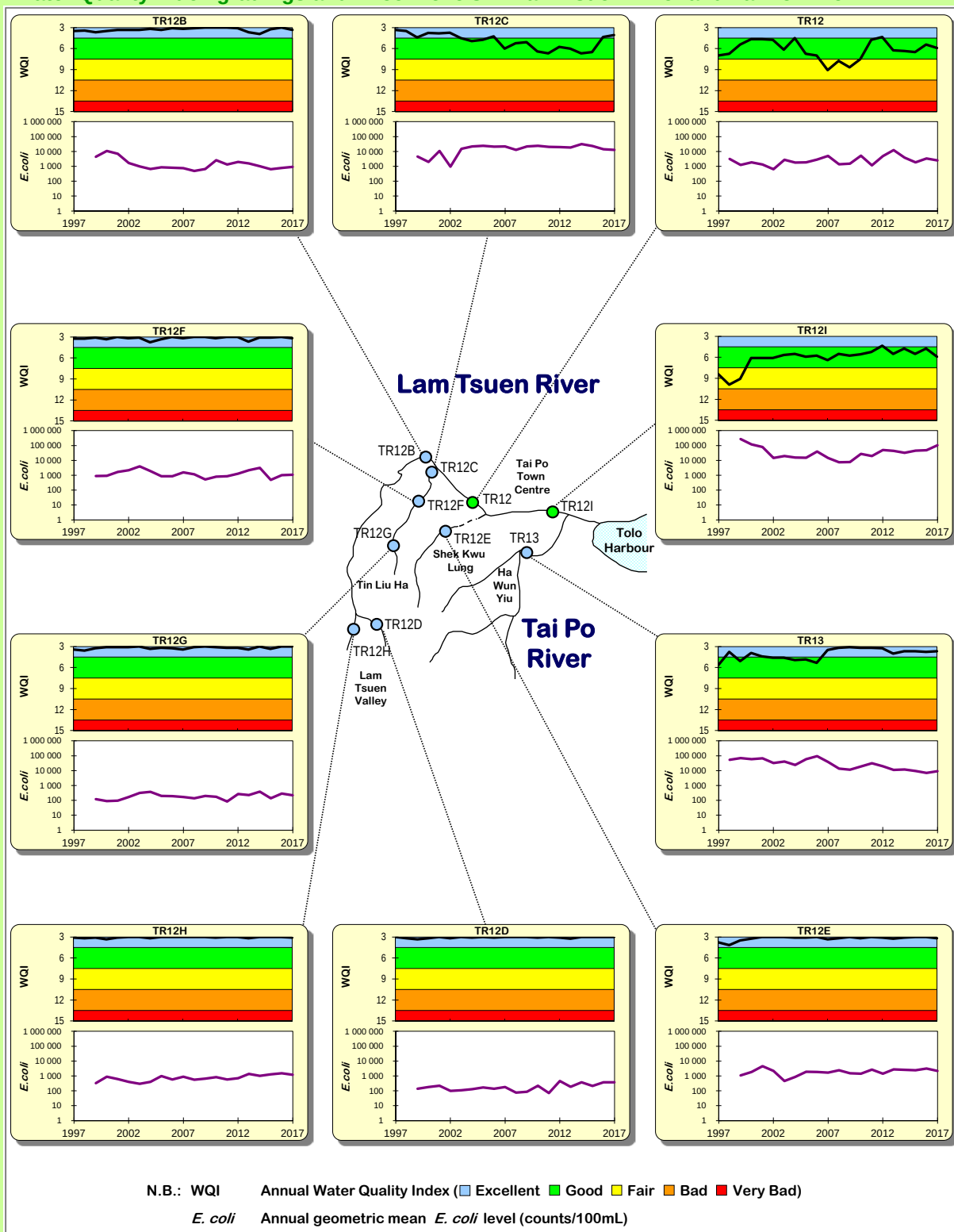
As for the WQI grading, 31 out of 32 (97%) of the river monitoring stations in the Eastern New Territories were graded “Good” or “Excellent” in 2017, same as in 2016. The remaining one station, i.e. Tseng Lan Shue Stream (JR3) at Tseng Lan Shue Tsuen, was in the “Fair” category. The overall improvement was due to the continued enforcement of the pollution control legislations, the implementation of Sewage Master Plans and extension of the public sewer to villages in the catchments.

The *E. coli* level at Shing Mun River Main Channel (TR19I) was considered as “Moderately low” in 2017, compared with “Moderate” in 2016.

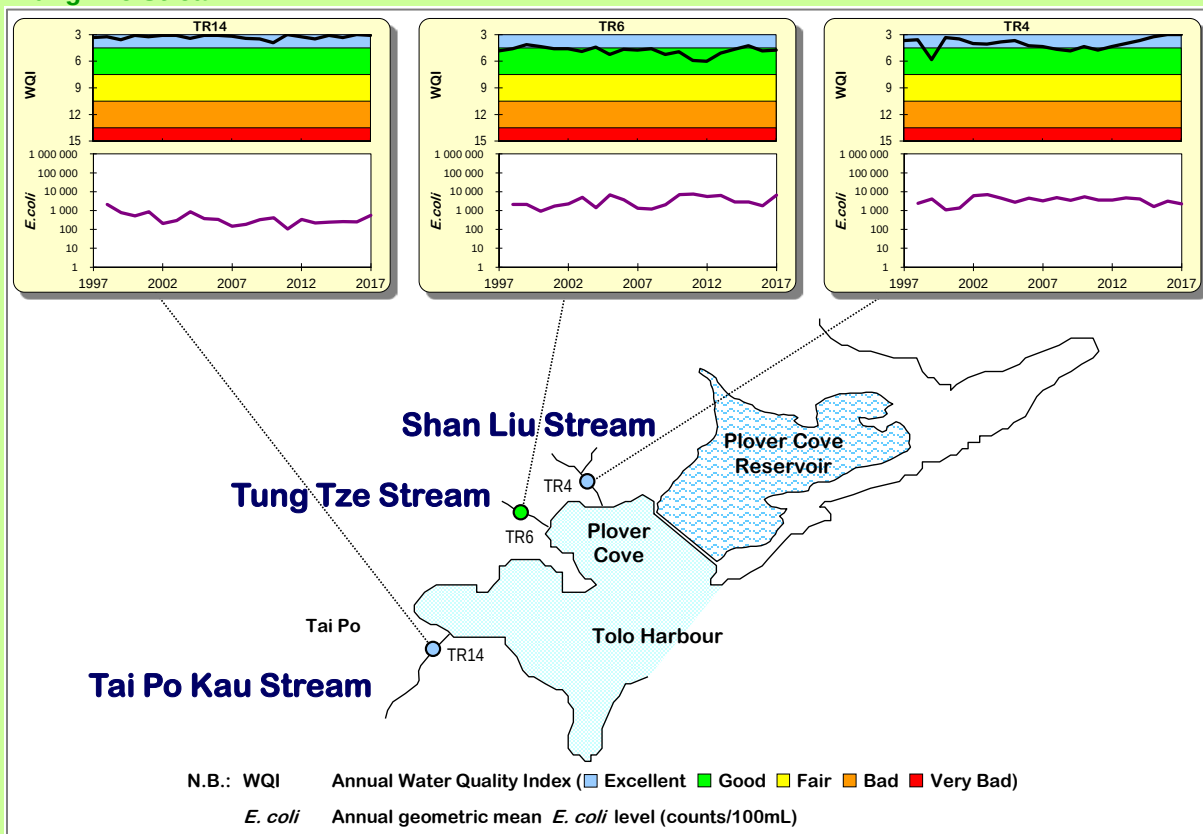
Water Quality Index gradings and *E. coli* levels in Shing Mun River



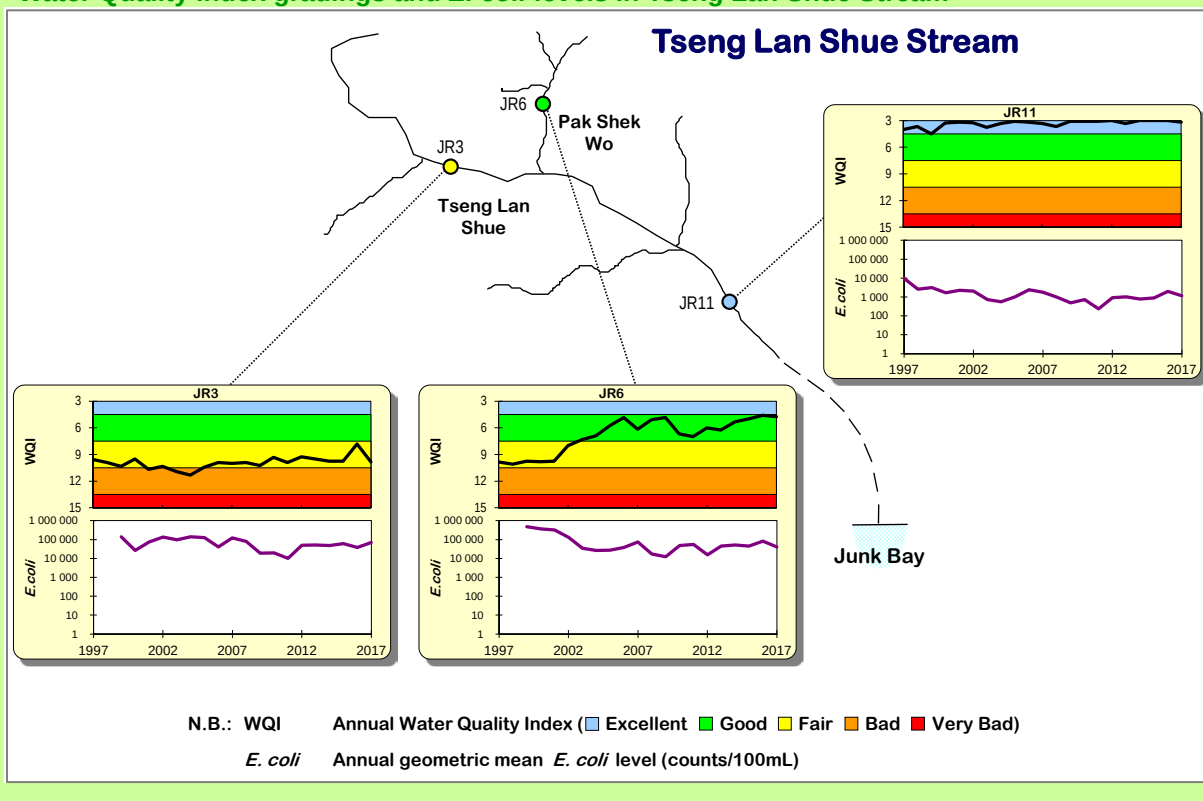
Water Quality Index gradings and *E. coli* levels in Lam Tsuen River and Tai Po River



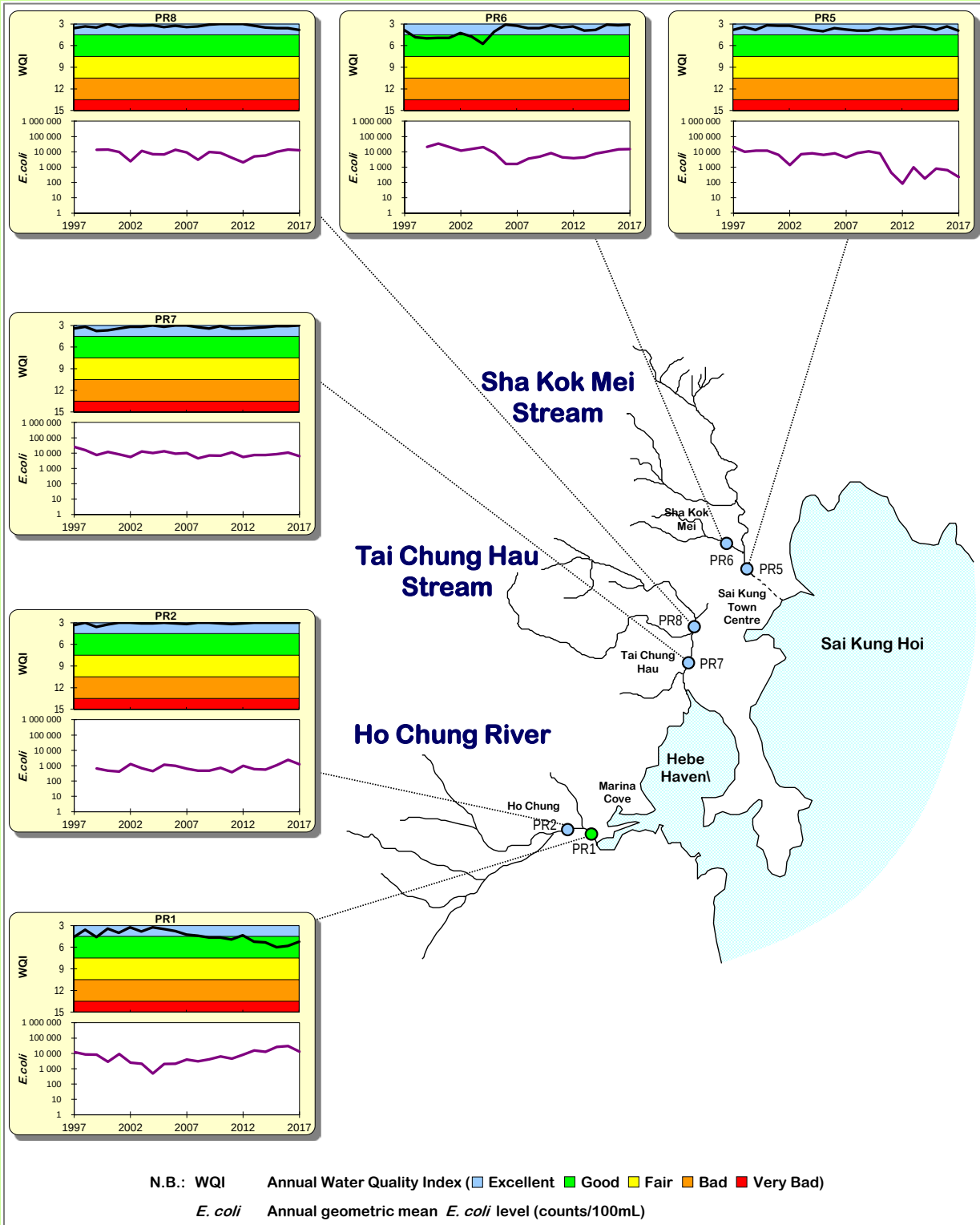
Water Quality Index gradings and *E. coli* levels in Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream



Water Quality Index gradings and *E. coli* levels in Tseng Lan Shue Stream



Water Quality Index gradings and *E. coli* levels in Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream



4. Northwestern New Territories rivers and streams

In the Northwestern New Territories, the EPD monitors a total of 13 rivers and streams in the Deep Bay WCZ, which flow into Shenzhen River or directly into Deep Bay (Shenzhen Bay). These include River Indus, Beas and Ganges in the North District, Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah in the Yuen Long District, and the six smaller streams located around Lau Fau Shan area.



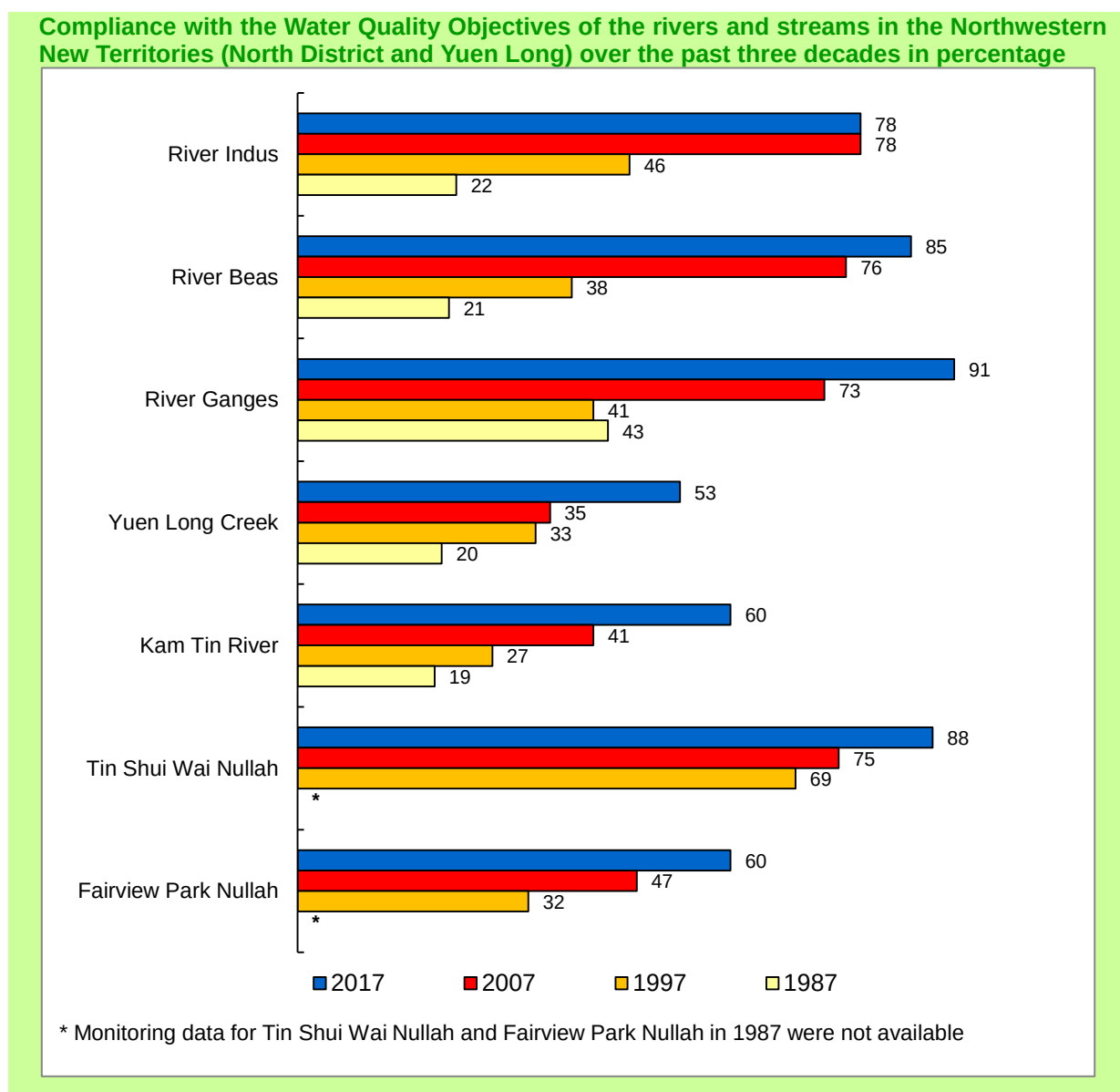
River Beas downstream

All the monitored rivers and streams in the Northwestern New Territories showed improvement in the WQO compliance rates over the past three decades. In 2017, the overall WQO compliance rate of all monitoring stations in the Northwestern New Territories was 80%, as compared with 70% in 2007, 55% in 1997 and 25% in 1987.

River Indus is a major river in the North District. It runs through rural areas like Lung Yeuk Tau, collects runoff from the densely populated Fanling and Sheung Shui urban areas, meets with River Beas before draining into Shenzhen River. The overall WQO compliance rate was 78% in 2017, as compared with 22% in 1987. In 2017, its downstream station (IN1) had a “Fair” grading in 2017, same as that in 2016. The mid-stream (IN2) and upstream (IN3) stations were “Good” and “Excellent” respectively, same as in 2016.

As a tributary of River Indus, River Beas recorded an overall WQO compliance rate of 85% in 2017, as compared with 21% in 1987. Individually, the upstream, mid-stream and downstream stations (RB1, RB2 and RB3) received “Excellent”, “Good” and “Good” grading respectively in 2017, similar to that in 2016.

For River Ganges, the overall WQO compliance rate was 91% in 2017, as compared with 43% in 1987. In terms of WQI, the downstream station GR1 and mid-stream station GR2 were graded “Good”, while the upstream station GR3 was graded “Excellent” in 2017.



As a major river in Yuen Long District, Yuen Long Creek passes through rural areas as well as the densely populated Yuen Long new town and Yuen Long Kau Hui, joins Kam Tin River before entering Deep Bay. In 2017, Yuen Long Creek’s overall compliance rate was 53%, compared with

35% in 2007 and 20% in 1987. For Kam Tin River, the overall compliance rate in 2017 was 60%. In terms of WQI, the two upstream stations (YL1 and YL2) recorded “Fair” grading, while the two downstream stations (YL3 and YL4) were graded “Bad” in 2017. Both stations in Kam Tin River (KT1 and KT2) also received WQI grading of “Bad” in 2017. The residual pollution load in Yuen Long and Kam Tin areas is largely contributed by unsewered villages. The Government is now carrying out a review of the local sewerage programme with a view to developing an action plan to improve the water quality in the above areas. Options including the provision of sewerage facilities for unsewered villages and possible installation of dry weather flow interceptors (DWFIs) where appropriate. Livestock farms are another key source of pollution in the areas. In addition to stepping up enforcement effect against illegal discharges, EPD has been collaborating with the Agriculture, Fisheries and Conservation Department in undertaking education programmes to tackle the pollution problems arising from livestock farms.

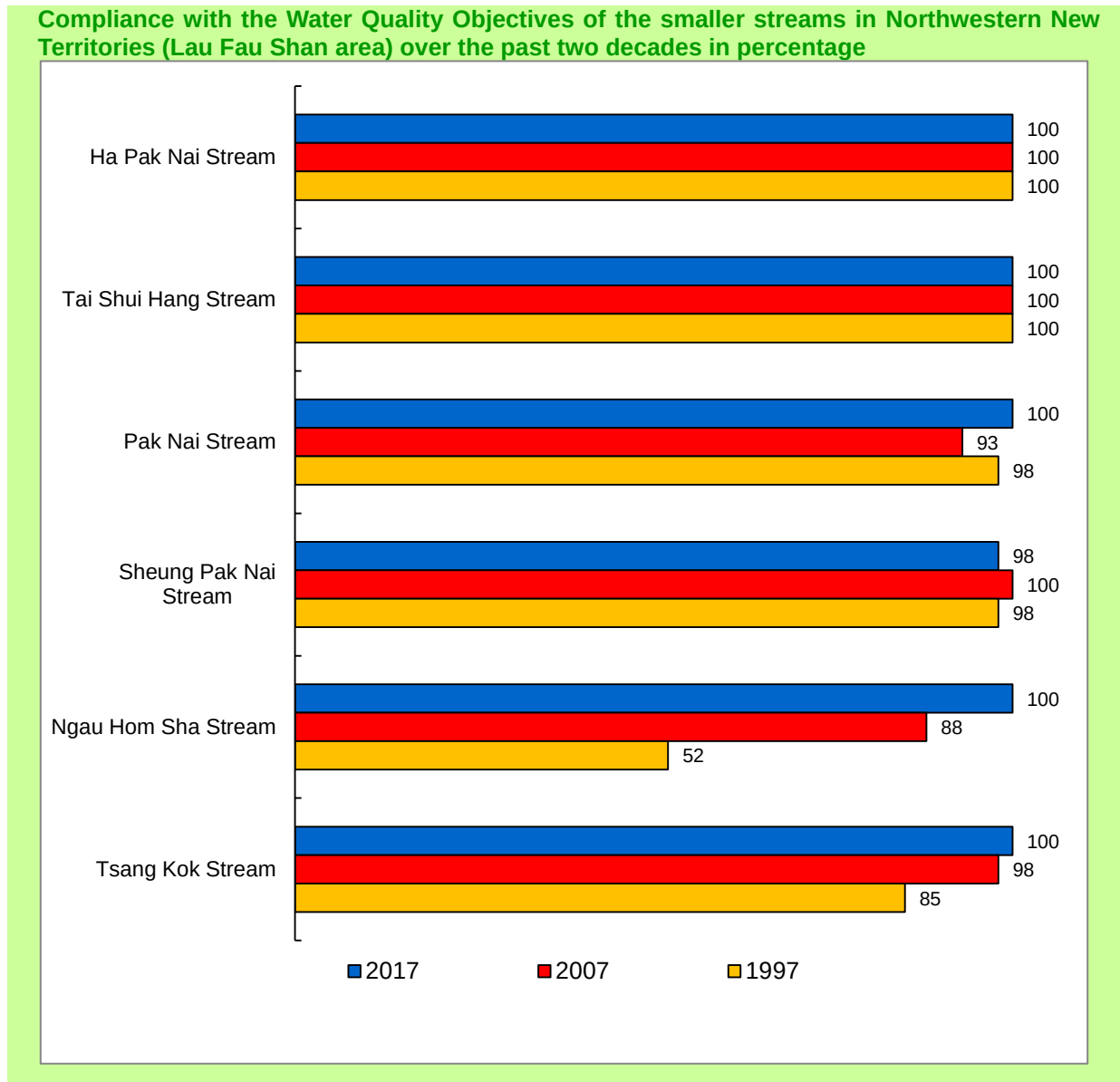


Kam Tin River

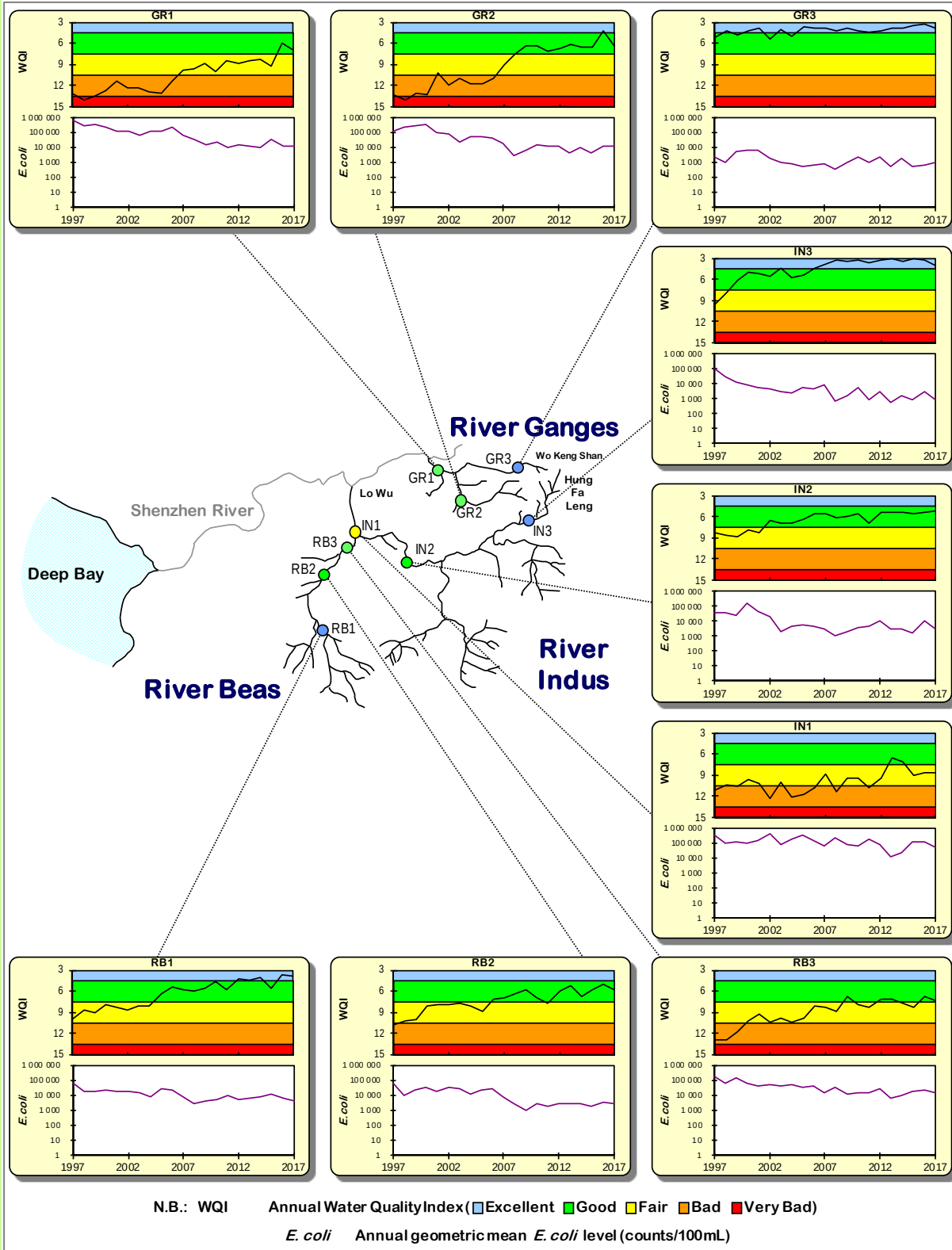
Tin Shui Wai Nullah had an overall WQO compliance of 88% in 2017 as compared with 75% in 2007 and 69% in 1997. In terms of WQI, its downstream station (TSR1) was graded “Fair” and its upstream station (TSR2) was graded “Good” in 2017.

The Fairview Park Nullah station (FVR1) recorded 60% WQO compliance in 2017 as compared with 47% in 2007 and 32% in 1997. In terms of WQI, it was graded “Fair” in 2017.

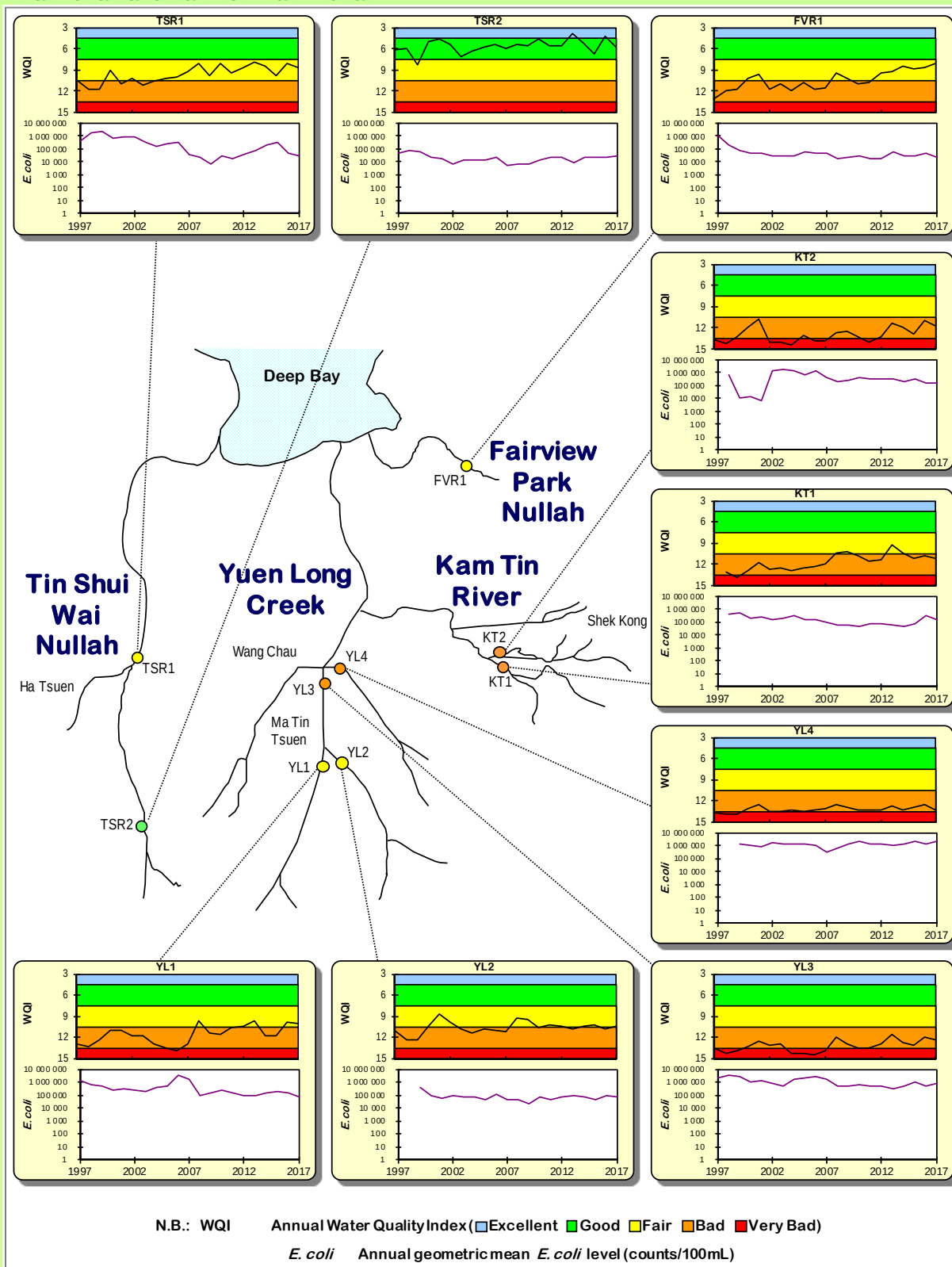
In 2017, the water quality of the six smaller streams around the Lau Fau Shan area remained good. Sheung Pak Nai Stream recorded an overall compliance of 98% while five of the remaining streams achieved full compliance. All streams had “Excellent” WQI grading.



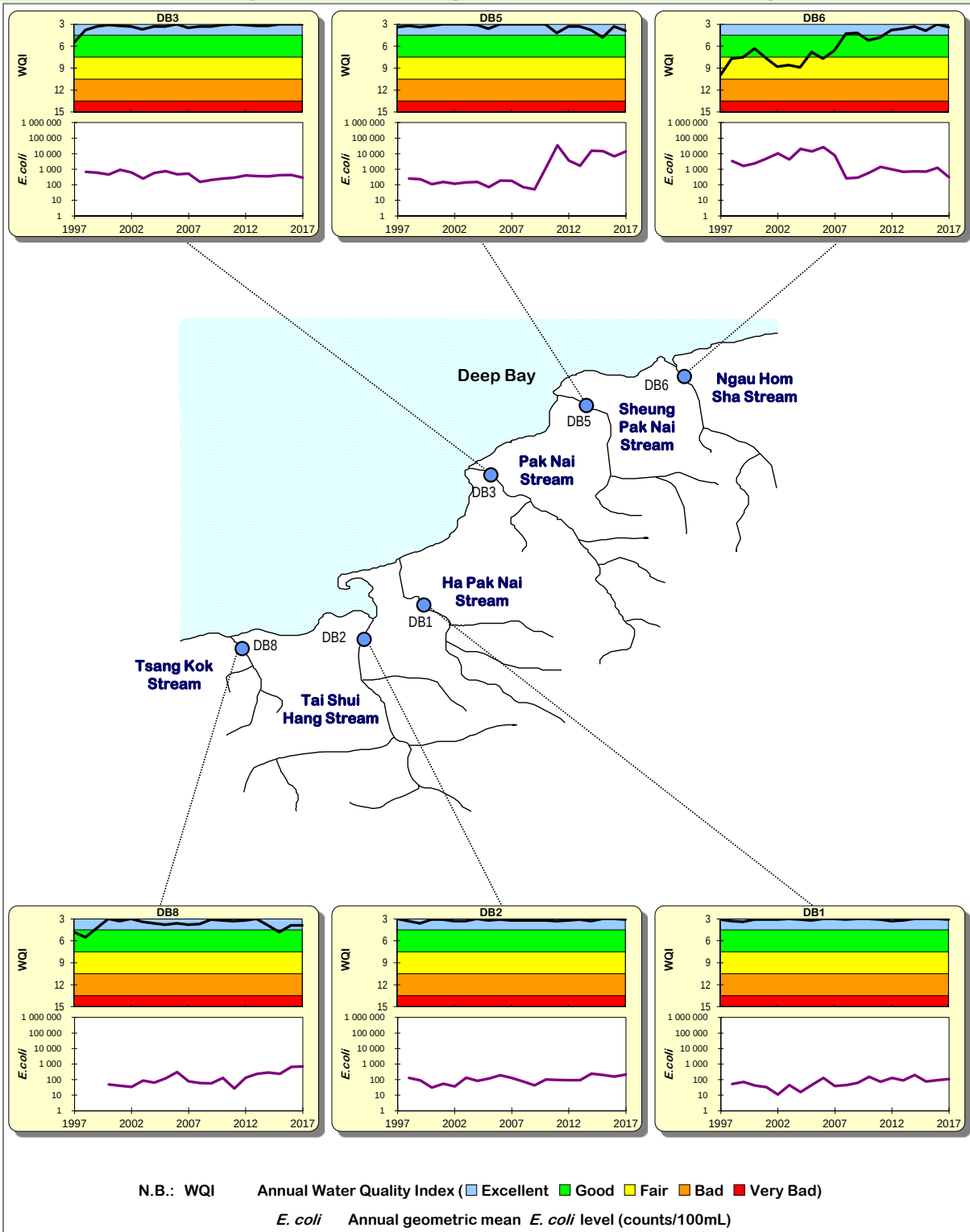
Water Quality Index gradings and *E. coli* levels in River Indus, River Beas and River Ganges



Water Quality Index gradings and *E. coli* levels in Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah



Water Quality Index gradings and *E. coli* levels in Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream and Tsang Kok Stream



5. Lantau Island rivers and streams

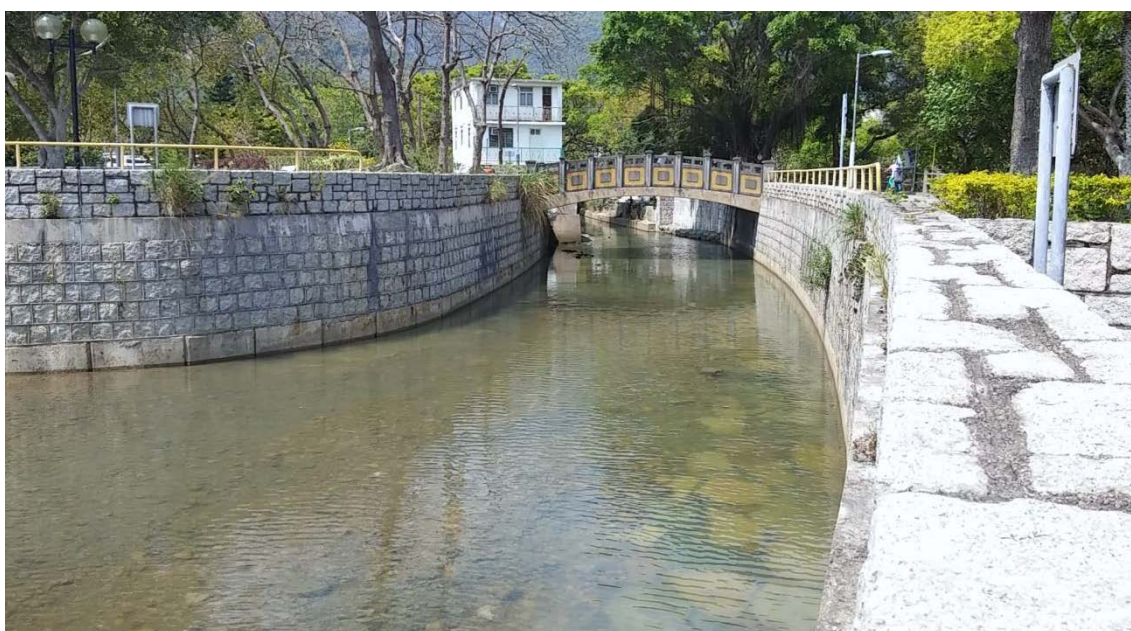
Lantau Island is large and less populated. Two rivers with a total of eight monitoring stations are monitored by the EPD: five along Mui Wo River on the south-eastern side of Lantau (Southern WCZ) and three at Tung Chung River on the north-western side of the island (North Western WCZ).

Overall, Mui Wo River and Tung Chung River generally displayed satisfactory water quality. The WQO compliance rates of Mui Wo River and Tung Chung River in 2017 were 99% and 96% respectively. As for the WQI grading, one monitoring station at Mui Wo River (MW5) and another at Tung Chung River (TC3) were graded “Good”. All of the remaining stations in both rivers were graded “Excellent” in 2017.

Compliance with the Water Quality Objectives of the rivers in Lantau Island over the past three decades in percentage

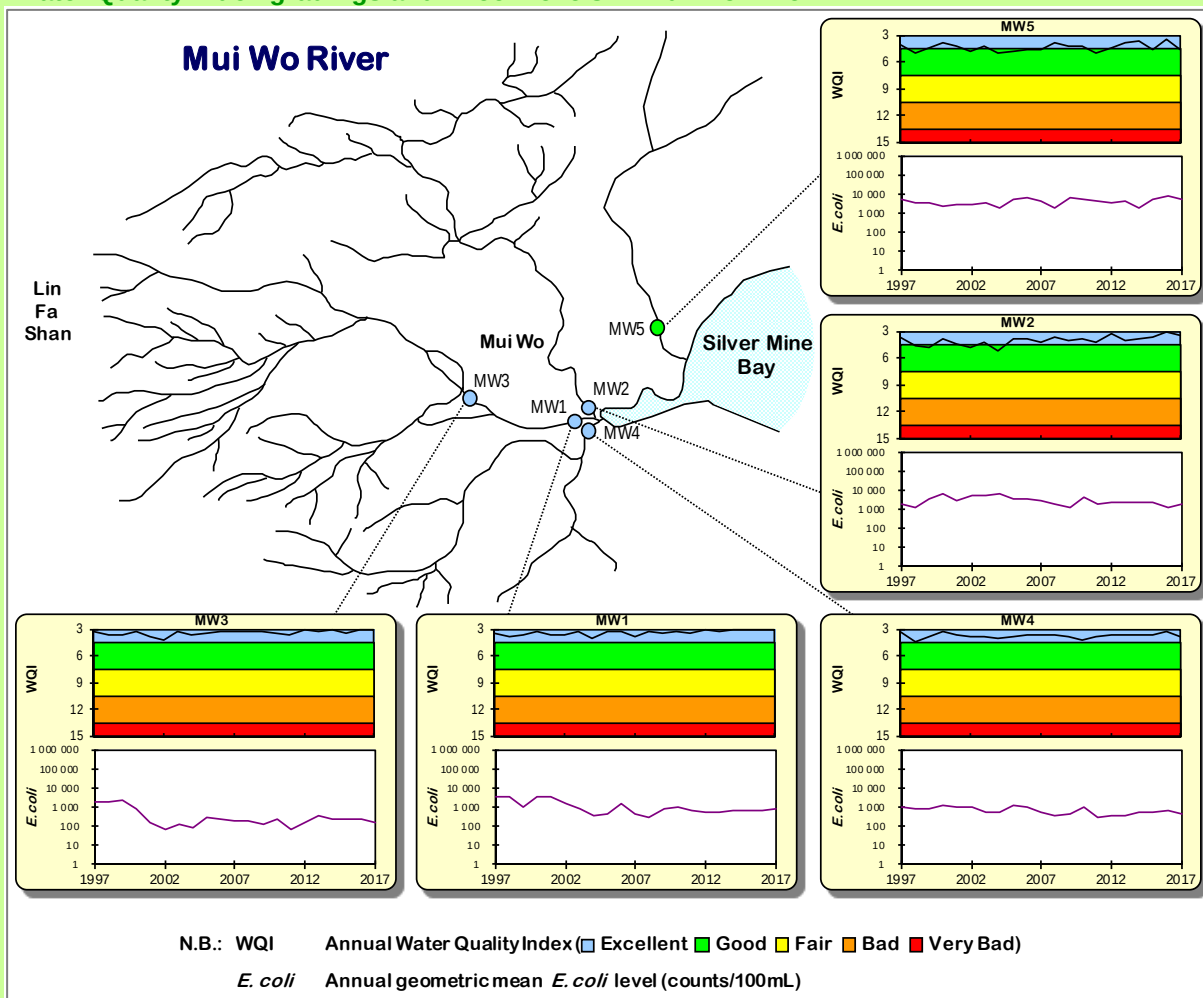


* Monitoring data for Tung Chung River in 1987 was not available

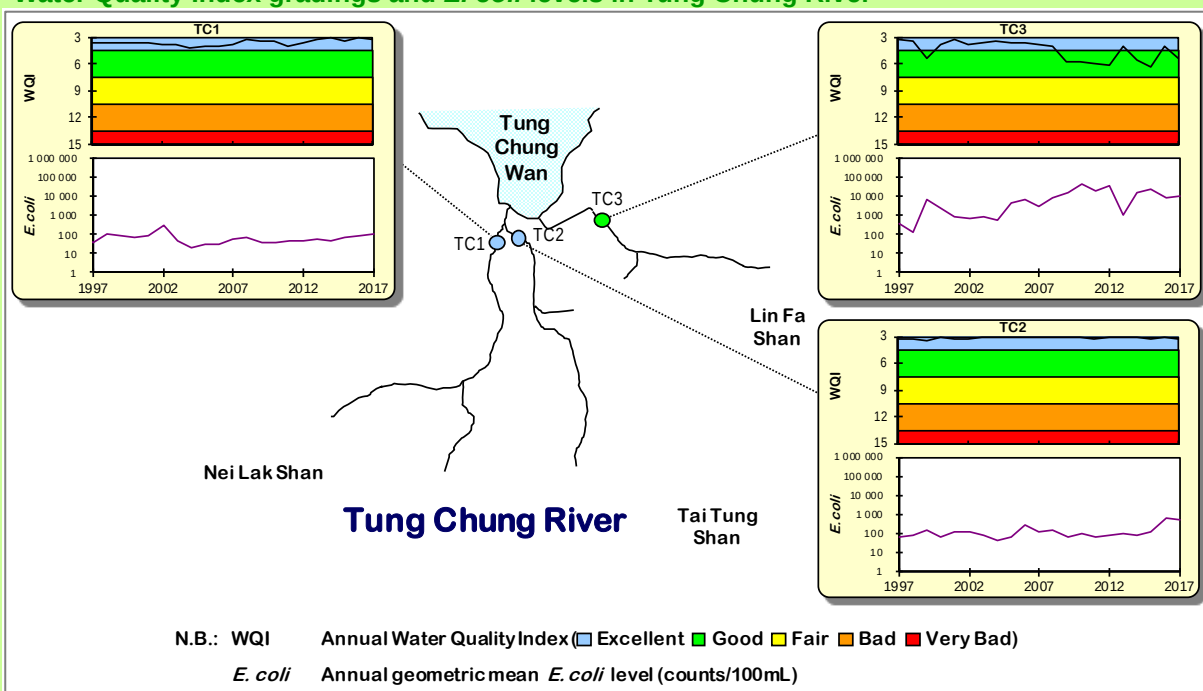


Mui Wo River

Water Quality Index gradings and *E. coli* levels in Mui Wo River



Water Quality Index gradings and *E. coli* levels in Tung Chung River



6. Southwestern New Territories and Kowloon rivers and streams

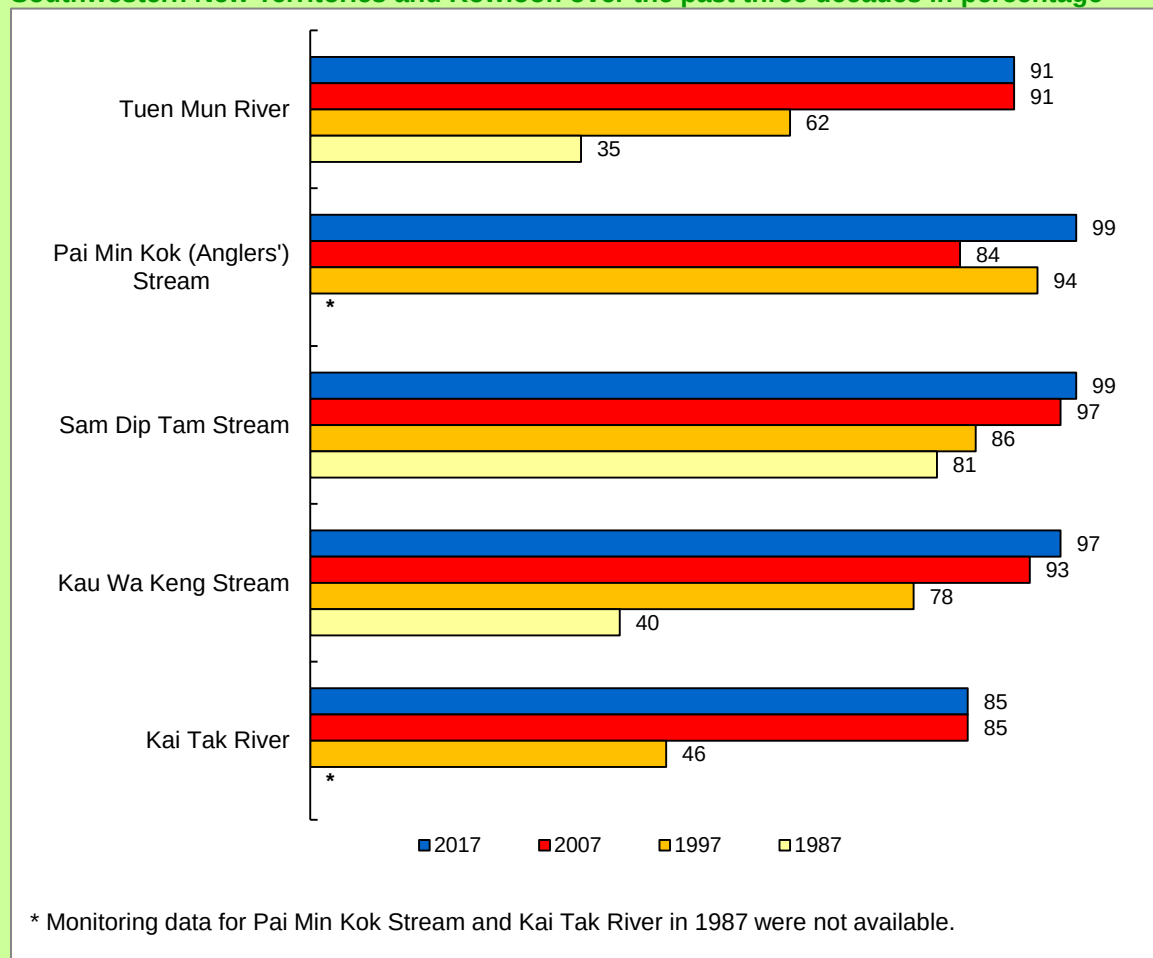
The Southwestern New Territories and Kowloon area covers Tuen Mun to the eastern end of Victoria Harbour. The five watercourses being monitored are Tuen Mun River in Tuen Mun (North Western WCZ), Pai Min Kok Stream near Sham Tseng (Western Buffer WCZ), Sam Dip Tam Stream near Tsuen Wan, Kau Wa Keng in Kwai Chung and Kai Tak River located in Kowloon (Victoria Harbour WCZ). There was substantial improvement in the water quality of these urban watercourses over the past three decades.



Tuen Mun River

In 2017, 17 of the 18 stations (94%) were graded “Excellent” or “Good”, as compared with 83% in 2016. Only one station was graded “Bad” this year (i.e. TN1 at Tuen Mun River upstream), as in 2016 and 2015. The overall WQO compliance rate of the rivers and streams in the area has risen from 75% in 1997 to 91% in 2017.

Compliance with the Water Quality Objectives of the rivers and streams in the Southwestern New Territories and Kowloon over the past three decades in percentage



As the major river in the Southwestern New Territories, Tuen Mun River passes through upstream areas like Lam Tei, San Hing Tsuen and Fu Tei, then the densely populated Tuen Mun town in its mid-stream, before entering Tuen Mun Typhoon Shelter. It showed marked improvement in the last three decades, with its WQO compliance rate rose steadily from 35% in 1987 to 91% in 2017. This improvement was attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun Sewage Master Plan. In 2017, the upstream station (TN1) was graded “Bad” mainly due to discharge from the unsewered areas. To prevent the pollutants from affecting Tuen Mun River, a DWFIs has been installed at West Rail Siu Hong Station to divert the flow at TN1 to the foul sewers and sewage treatment works for treatment. All other five stations (TN2, TN3, TN4, TN5 and TN6) were graded “Good” or “Excellent”.

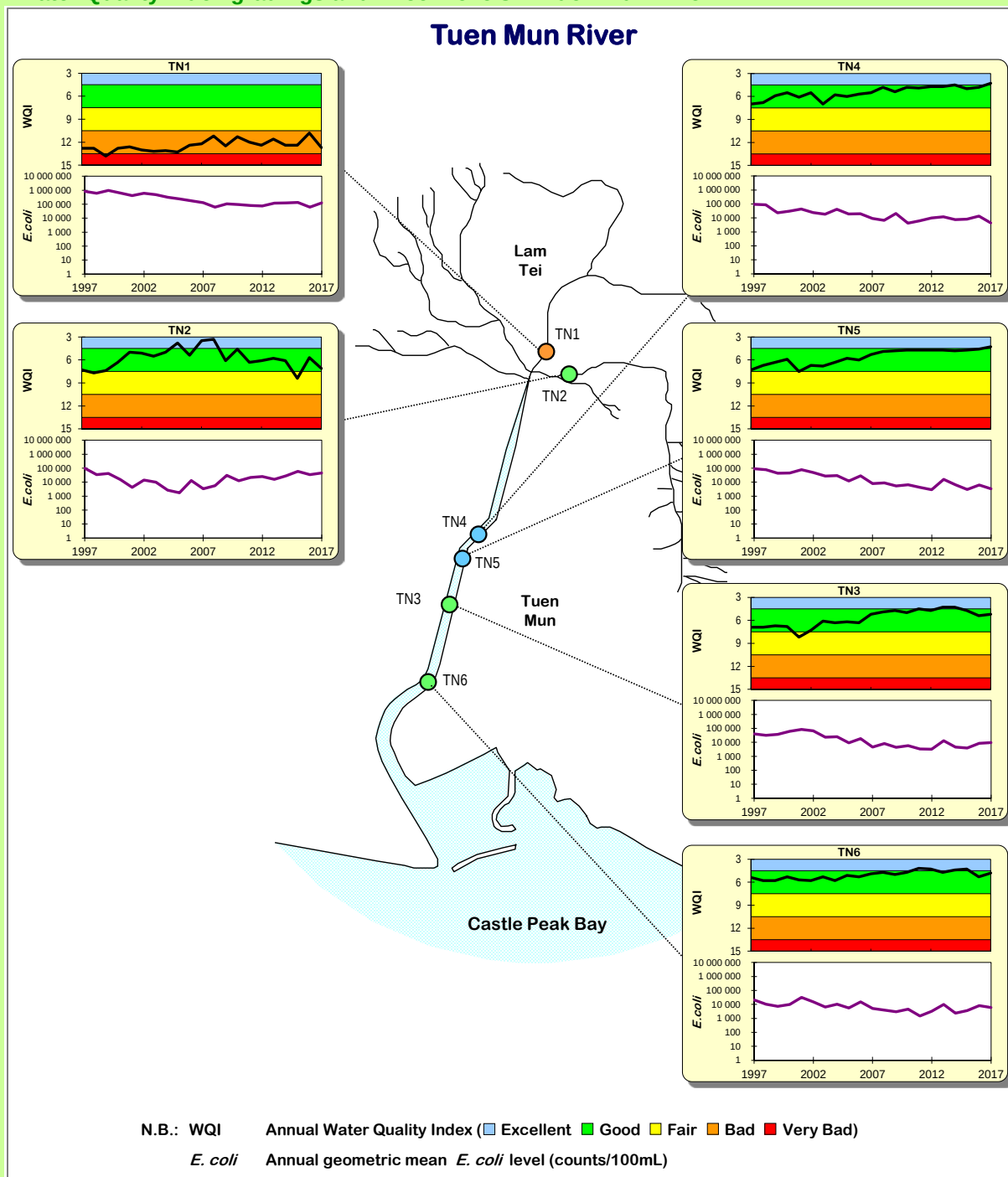
Pai Min Kok Stream achieved an overall WQO compliance rate of 99% in 2017. In terms of WQI, both its upstream station (AN2) and downstream station (AN1) were graded “Excellent” in 2017.

Sam Dip Tam Stream of Tsuen Wan also achieved 99% compliance with the WQO and all its three monitoring stations were graded “Excellent” in 2017.

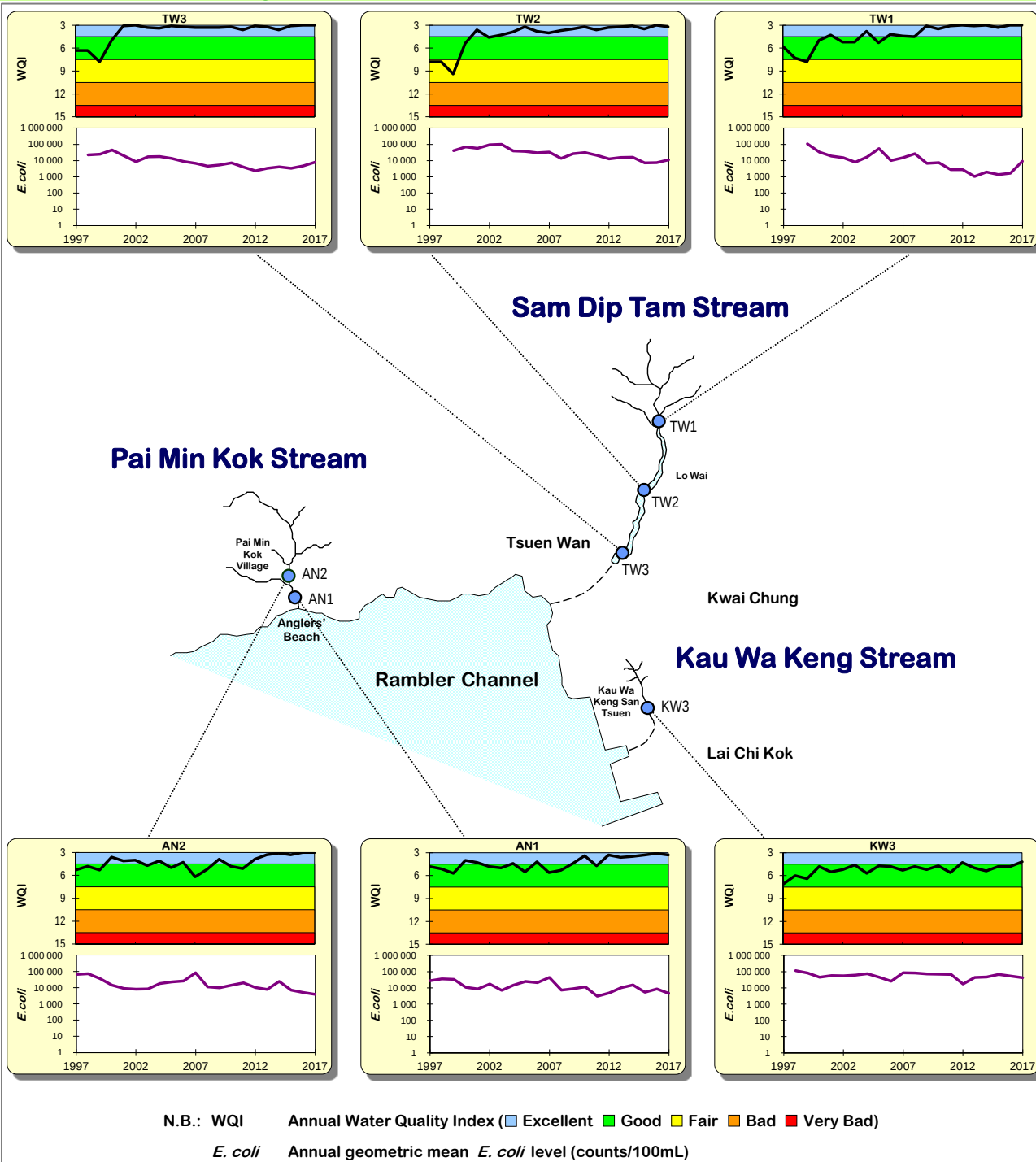
The monitoring station KW3 at Kau Wa Keng Stream in Kwai Chung achieved a WQO compliance rate of 97% in 2017, as compared with 40% in 1987. Its WQI grading was “Excellent” in 2017.

Kai Tak River had a WQO compliance rate of 85% in 2017, as compared with 46% in 1997. All six monitoring stations in Kai Tak River were grade “Good” in 2017, compared with two downstream stations being graded as “Fair” while the other four upstream stations being graded as “Good” in 2016. It was expected that the water quality of Kai Tak River would further improve once Civil Engineering and Development Department completed its improvement work in the Kai Tak Development Area and Drainage Services Department repaired and upgraded the sewer system in its catchment area.

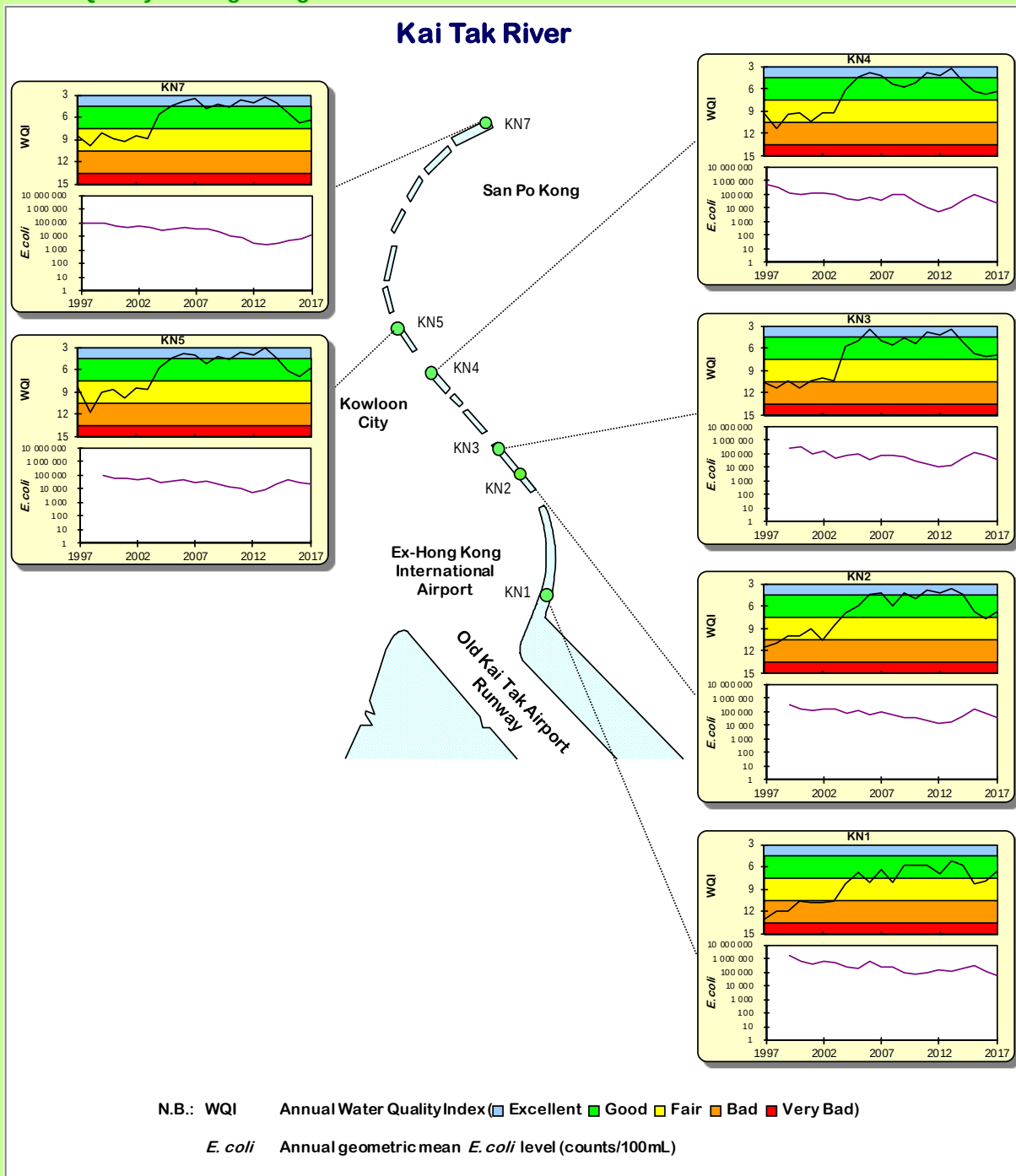
Water Quality Index gradings and *E. coli* levels in Tuen Mun River



Water Quality Index gradings and *E. coli* levels in Pai Min Kok (Anglers') Stream, Sam Dip Tam Stream and Kau Wa Keng Stream



Water Quality Index gradings and *E. coli* levels in Kai Tak River



Appendices

Summary of river water quality monitoring stations and sampling frequencies in 2017

Area	Watercourse	Monitoring Station (Number)	Sampling frequency
Eastern New Territories			
Sha Tin	Shing Mun River	TR19I (1)	monthly
	<i>Shing Mun Main Channel</i>	TR23A, TR23L (2)	monthly
	<i>Siu Lek Yuen Nullah</i>	TR17, TR17L (2)	monthly
	<i>Fo Tan Nullah</i>	KY1 (1)	monthly
	<i>Kwun Yam Shan Stream</i>	TR19, TR19A, TR19C (3)	monthly
	<i>Tai Wai Nullah</i>	TR20B (1)	monthly
Tai Po Town Centre	Lam Tsuen River	TR12, TR12B, TR12C, TR12D, TR12E, TR12F, TR12G, TR12H, TR12I (9)	monthly
	Tai Po River	TR13 (1)	monthly
Tai Po Rural Area	Tai Po Kau Stream	TR14 (1)	monthly
	Shan Liu Stream	TR4 (1)	monthly
	Tung Tze Stream	TR6 (1)	monthly
Sai Kung	Ho Chung River	PR1, PR2 (2)	monthly
	Sha Kok Mei Stream	PR5, PR6 (2)	monthly
	Tai Chung Hau Stream	PR7, PR8 (2)	monthly
Tseung Kwan O	Tseng Lan Shue Stream	JR3, JR6, JR11 (3)	monthly
Northwestern New Territories			
North District	River Indus (Ng Tung River)	IN1, IN2, IN3 (3)	monthly
	River Beas (Sheung Yue River)	RB1, RB2, RB3 (3)	monthly
	River Ganges (Ping Yuen River)	GR1, GR2, GR3 (3)	monthly
Yuen Long	Yuen Long Creek	YL1, YL2, YL3, YL4 (4)	monthly
	Kam Tin River	KT1, KT2 (2)	monthly
	Tin Shui Wai Nullah	TSR1, TSR2 (2)	monthly
	Fairview Park Nullah	FVR1 (1)	monthly
Lau Fau Shan	Ha Pak Nai Stream	DB1 (1)	monthly
	Tai Shui Hang Stream	DB2 (1)	monthly
	Pak Nai Stream	DB3 (1)	monthly
	Sheung Pak Nai Stream	DB5 (1)	monthly
	Ngau Hom Sha Stream	DB6 (1)	monthly
	Tsang Kok Stream	DB8 (1)	monthly
Lantau Island			
Mui Wo	Mui Wo River	MW1, MW2, MW3, MW4, MW5 (5)	monthly
Tung Chung	Tung Chung River	TC1, TC2, TC3 (3)	monthly
Southwestern New Territories and Kowloon			
Tuen Mun	Tuen Mun River	TN1, TN2, TN3, TN4, TN5, TN6 (6)	monthly
Tsuen Wan and Kwai Chung	Pai Min Kok (Anglers') Stream	AN1, AN2 (2)	monthly
	Sam Dip Tam Stream	TW1, TW2, TW3 (3)	monthly
	Kau Wa Keng Stream	KW3 (1)	monthly
Kowloon	Kai Tak River	KN1, KN2, KN3, KN4, KN5, KN7 (6)	monthly
Total	30	82	-

River water quality parameters and the methods of analysis (part 1 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Physical Chemical Properties		
Water temperature	0.1 °C	Multi-parameter water quality data logger, model YSI-6820 / On-site measurement / EPD
Dissolved Oxygen	0.1 mg/L, 1%	
pH	0.1	
Conductivity	1 µS/cm	
Turbidity	0.1 NTU	
Flow	1 L/s	Electromagnetic flow meter, model Flo-mate 2000 / Global Water Flow Probe, model FP211 / On-site measurement / EPD
Solid Contents		
Suspended Solids	0.5 mg/L	In-house method GL-PH-23, based on APHA ² 22ed 2540 D / Government Laboratory
Total Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 22ed 2540 B & E / Government Laboratory
Total Volatile Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 22ed 2540 B & E / Government Laboratory
Aggregate Organics		
5-Day Biochemical Oxygen Demand (BOD ₅)	0.1 mg/L	In-house method based on APHA 18ed 5210 B / EPD
Chemical Oxygen Demand (COD)	2 mg/L	In-house method GL-OR-38 & GL-OR-39, based on ASTM ³ D1252-06, Method B or APHA 22ed 5220 C & D / Government Laboratory
Total Organic Carbon (TOC)	1 mg/L	In-house method GL-OR-32, based on APHA 20ed 5310 B / Government Laboratory
Faecal Bacteria		
<i>E. coli</i>	1 count / 100 mL	In-house method ⁴ , membrane filtration with CHROMagar Liquid ECC medium / EPD
Faecal Coliforms	1 count / 100 mL	
Nutrients		
Ammonia-Nitrogen	0.005 mg/L	In-house method GL-IN-15, based on APHA 22ed 4500-NH ₃ / Government Laboratory
Nitrite-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₂ ⁻ B (FIA) / Government Laboratory
Nitrate-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₃ ⁻ F & I (FIA) / Government Laboratory
Total Kjeldahl Nitrogen	0.05 mg/L	In-house method GL-IN-14 & GL-IN-15, based on ASTM D3590-11 B (FIA) & APHA 20ed 4500-N A&D (FIA) / Government Laboratory
Ortho-phosphate Phosphorus	0.002 mg/L	In-house method GL-IN-16, based on ASTM D515-88 A (FIA) / Government Laboratory
Total Phosphorus	0.02 mg/L	In-house method GL-IN-14 & GL-IN-16, based on ASTM D515-88 B (FIA) & APHA 20ed 4500-P G (FIA) / Government Laboratory
Molybdate-Reactive Silica	0.05 mg/L	In-house method GL-IN-17, based on APHA 20ed 4500-SiO ₂ C&E (FIA) / Government Laboratory

Reference notes:

1. Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.
2. APHA - American Public Health Association: *Standard Methods for the Examination of Water and Wastewater*.
3. ASTM - *Annual Book of American Society for the Testing and Materials Standards*, Vol. 11.01 & 11.02.
4. i) Ho, B.S.W. and Tam, T.Y. (1997). Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. *Wat. Sci. Tech.*, **35**, 409-413.
ii) DoE and DHSS (1983). "The bacteriological examination of drinking water supplies 1982. Report on Public Health and Medical Subjects No. 71. Methods for the Examination of Waters and Associated Materials". Department of Environment, Department of Health and Social Security, Public Health Laboratory Service, H.M.S.O. London.

River water quality parameters and the methods of analysis (part 2 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Metals		
Aluminium	50 µg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Antimony	1 µg/L	
Arsenic	1 µg/L	
Barium	1 µg/L	
Beryllium	1 µg/L	
Boron	50 µg/L	
Cadmium	0.1 µg/L	
Chromium	1 µg/L	
Copper	1 µg/L	
Iron	50 µg/L	
Lead	1 µg/L	
Manganese	10 µg/L	
Mercury	1 µg/L	
Molybdenum	2 µg/L	
Nickel	1 µg/L	
Silver	1 µg/L	
Thallium	1 µg/L	
Vanadium	2 µg/L	
Zinc	10 µg/L	
Industrial and Commercial Pollutants		
Cyanide	0.01 mg/L	In-house method GL-IN-42, based on ASTM D 2036-09 or APHA 22ed 4500-CN / Government Laboratory
Fluoride	0.2 mg/L	In-house method GL-IN-47, based on APHA 22ed 4500-F ⁻ C & G (Ion Selective Electrode) and ASTM D1179-99 B (FIA) / Government Laboratory
Anionic Surfactants (as Manoxol OT)	0.05 mg/L	In-house method GL-OR-30, based on BS 6068, Section 2.23 (1994) (Colorimetric) & In-house method GL-OR-65, based on Abbott, D.C. “Analyst”, Vol.87, p.286(1962) & S. Motomizu et al., “Analyst” Vol.113, p.747(1988) (FIA) / Government Laboratory
Oil and Grease	0.5 mg/L	In-house method GL-OR-26, based on APHA 22ed 5520 B / Government Laboratory
Sulphide Contents		
Free Hydrogen Sulphide	0.01 mg/L	In-house method GL-IN-46, based on APHA 22ed 4500S ²⁻ D (Colorimetric) / Government Laboratory
Sulphide	0.02 mg/L	
Plant Pigments		
Chlorophyll-a	0.2 µg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 (spectrophotometric) / Government Laboratory
Pheo-Pigment	0.2 µg/L	

Key Water Quality Objectives (WQOs) for river monitoring stations in the Eastern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Tolo Harbour and Channel Water Control Zone						
Shing Mun River	KY1	6.5 - 8.5	3	15	20	4
	TR17	6.5 - 8.5	5	30	20	4
	TR17L	6.5 - 8.5	5	30	20	4
	TR19	6.5 - 8.5	5	30	20	4
	TR19A	6.5 - 8.5	5	30	20	4
	TR19C	6.5 - 8.5	5	30	20	4
	TR19I	6.0 - 9.0	5	30	25	4
	TR20B	6.5 - 8.5	5	30	20	4
	TR23A	6.5 - 8.5	3	15	20	4
	TR23L	6.5 - 8.5	3	15	20	4
Lam Tsuen River	TR12	6.5 - 8.5	3	15	20	4
	TR12B	6.5 - 8.5	3	15	20	4
	TR12C	6.5 - 8.5	3	15	20	4
	TR12D	6.5 - 8.5	3	15	20	4
	TR12E	6.5 - 8.5	3	15	20	4
	TR12F	6.5 - 8.5	3	15	20	4
	TR12G	6.5 - 8.5	3	15	20	4
	TR12H	6.5 - 8.5	3	15	20	4
	TR12I	6.0 - 9.0	5	30	25	4
Tai Po River	TR13	6.5 - 8.5	5	30	20	4
Tai Po Kau Stream	TR14	6.0 - 9.0	5	30	25	4
Shan Liu Stream	TR4	6.0 - 9.0	5	30	25	4
Tung Tze Stream	TR6	6.0 - 9.0	5	30	25	4
Port Shelter Water Control Zone						
Ho Chung River	PR1	6.5 - 8.5	5	30	25	4
	PR2	6.5 - 8.5	5	30	25	4
Sha Kok Mei Stream	PR5	6.0 - 9.0	5	30	25	4
	PR6	6.0 - 9.0	5	30	25	4
Tai Chung Hau Stream	PR7	6.0 - 9.0	5	30	25	4
	PR8	6.0 - 9.0	5	30	25	4
Junk Bay Water Control Zone						
Tseng Lan Shue Stream	JR3	6.0 - 9.0	5	30	25	4
	JR6	6.0 - 9.0	5	30	25	4
	JR11	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Northwestern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Deep Bay Water Control Zone						
River Indus	IN1	6.5 - 8.5	3	15	20	4
	IN2	6.5 - 8.5	3	15	20	4
	IN3	6.5 - 8.5	3	15	20	4
River Beas	RB1	6.5 - 8.5	3	15	20	4
	RB2	6.5 - 8.5	3	15	20	4
	RB3	6.5 - 8.5	3	15	20	4
River Ganges	GR1	6.5 - 8.5	3	15	20	4
	GR2	6.5 - 8.5	3	15	20	4
	GR3	6.5 - 8.5	3	15	20	4
Yuen Long Creek	YL1	6.5 - 8.5	3	15	20	4
	YL2	6.5 - 8.5	3	15	20	4
	YL3	6.5 - 8.5	5	30	20	4
	YL4	6.5 - 8.5	5	30	20	4
Kam Tin River	KT1	6.5 - 8.5	3	15	20	4
	KT2	6.5 - 8.5	3	15	20	4
Tin Shui Wai Nullah	TSR1	6.0 - 9.0	5	30	20	4
	TSR2	6.0 - 9.0	5	30	20	4
Fairview Park Nullah	FVR1	6.0 - 9.0	5	30	20	4
Ha Pak Nai Stream	DB1	6.0 - 9.0	5	30	20	4
Tai Shui Hang Stream	DB2	6.0 - 9.0	5	30	20	4
Pak Nai Stream	DB3	6.0 - 9.0	5	30	20	4
Sheung Pak Nai Stream	DB5	6.0 - 9.0	5	30	20	4
Ngau Hom Sha Stream	DB6	6.0 - 9.0	5	30	20	4
Tsang Kok Stream	DB8	6.0 - 9.0	5	30	20	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations on Lantau Island

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Southern Water Control Zone						
Mui Wo River	MW1	6.5 - 8.5	5	30	20	4
	MW2	6.5 - 8.5	5	30	20	4
	MW3	6.5 - 8.5	5	30	20	4
	MW4	6.5 - 8.5	5	30	20	4
	MW5	6.0 - 9.0	5	30	25	4
North Western Water Control Zone						
Tung Chung River	TC1	6.0 - 9.0	5	30	25	4
	TC2	6.0 - 9.0	5	30	25	4
	TC3	6.0 - 9.0	5	30	25	4

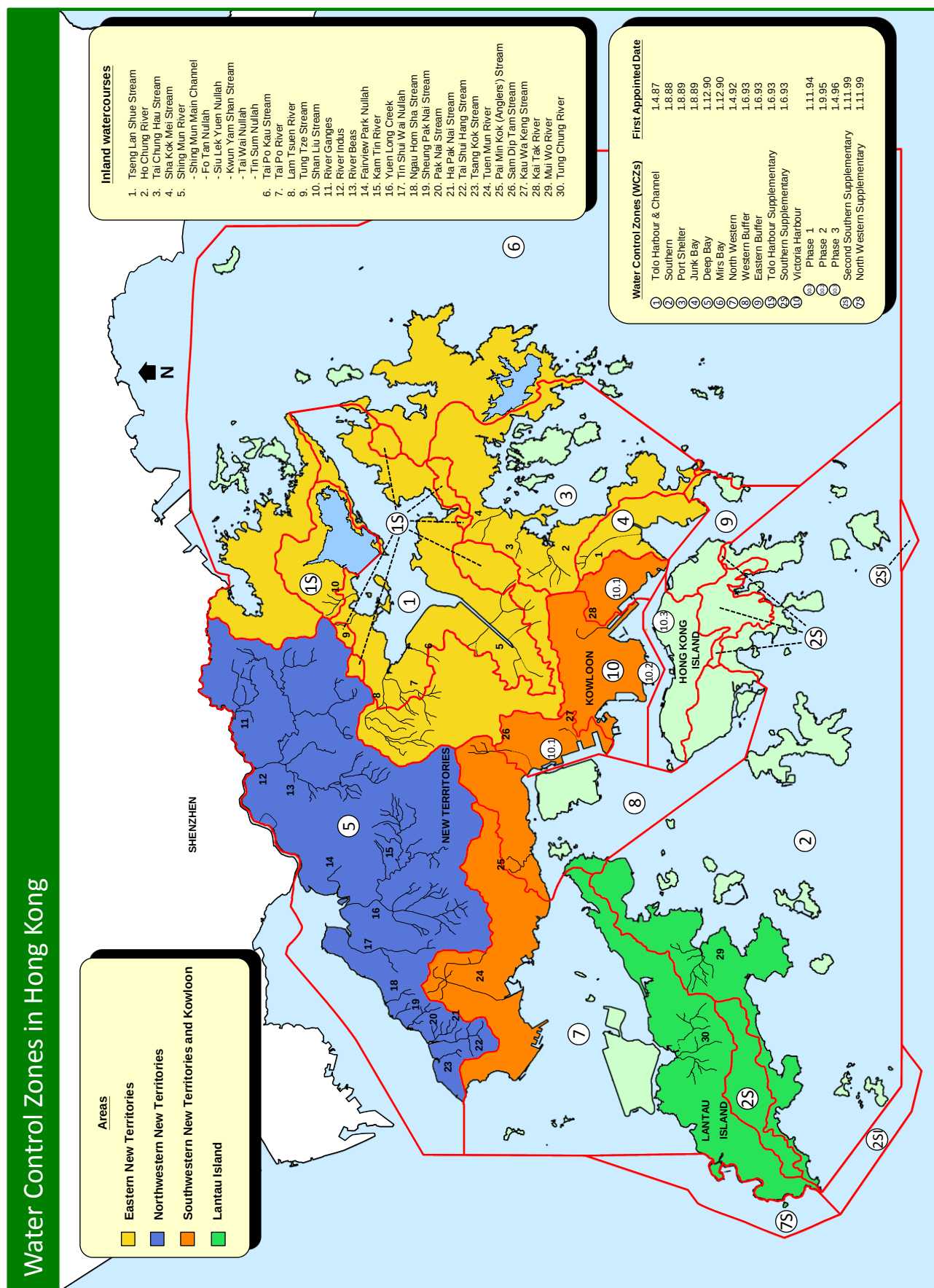
* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Southwestern New Territories and Kowloon

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
North Western Water Control Zone						
Tuen Mun River	TN1	6.0 - 9.0	5	30	25	4
	TN2	6.5 - 8.5	3	15	20	4
	TN3	6.0 - 9.0	5	30	25	4
	TN4	6.0 - 9.0	5	30	25	4
	TN5	6.0 - 9.0	5	30	25	4
	TN6	6.0 - 9.0	5	30	25	4
Western Buffer Water Control Zone						
Pai Min Kok (Anglers') Stream	AN1	6.0 - 9.0	5	30	25	4
	AN2	6.0 - 9.0	5	30	25	4
Victoria Harbour Water Control Zone						
Sam Dip Tam Stream	TW1	6.0 - 9.0	5	30	25	4
	TW2	6.0 - 9.0	5	30	25	4
	TW3	6.0 - 9.0	5	30	25	4
Kau Wa Keng Stream	KW3	6.0 - 9.0	5	30	25	4
Kai Tak River	KN1	6.0 - 9.0	5	30	25	4
	KN2	6.0 - 9.0	5	30	25	4
	KN3	6.0 - 9.0	5	30	25	4
	KN4	6.0 - 9.0	5	30	25	4
	KN5	6.0 - 9.0	5	30	25	4
	KN7	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Water Control Zones in Hong Kong



Summary of water quality monitoring data for Shing Mun River (Main Channel and Siu Lek Yuen Nullah) in 2017

Parameter	Unit	Shing Mun Main Channel	Siu Lek Yuen Nullah	
		TR19I	TR23L	TR23A
Dissolved oxygen	mg/L	7.3 (5.6 - 8.2)	8.8 (7.9 - 12.5)	6.8 (5.9 - 8.7)
pH		8.1 (7.9 - 8.2)	8.5 (7.9 - 9.3)	7.6 (7.4 - 8.0)
Suspended solids	mg/L	5.0 (1.8 - 22.0)	2.2 (0.9 - 21.0)	4.4 (1.9 - 11.0)
5-day Biochemical Oxygen Demand	mg/L	2.9 (1.7 - 6.0)	1.9 (0.6 - 7.5)	1.4 (0.9 - 2.8)
Chemical Oxygen Demand	mg/L	13 (5 - 20)	5 (<2 - 11)	10 (4 - 20)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	800 (40 - 15 000)	3 600 (260 - 24 000)	2 800 (570 - 8 000)
Faecal coliforms	counts/ 100mL	5 000 (970 - 39 000)	53 000 (1 500 - 570 000)	17 000 (2 400 - 140 000)
Ammonia-nitrogen	mg/L	0.115 (0.045 - 0.280)	0.015 (<0.005 - 0.094)	0.245 (0.120 - 0.350)
Nitrate-nitrogen	mg/L	0.130 (0.038 - 0.400)	0.175 (<0.002 - 0.730)	0.340 (0.098 - 0.460)
Total Kjeldahl Nitrogen	mg/L	0.38 (0.30 - 0.72)	0.19 (0.11 - 0.72)	0.42 (0.22 - 0.52)
Ortho-phosphate Phosphorus	mg/L	<0.002 (<0.002 - 0.010)	0.005 (0.002 - 0.042)	0.017 (0.003 - 0.032)
Total phosphorus	mg/L	0.05 (0.03 - 0.12)	<0.02 (<0.02 - 0.09)	0.05 (0.04 - 0.10)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	51 (<50 - 275)	72 (<50 - 293)	66 (<50 - 129)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 3)	<1 (<1 - 1)	1 (<1 - 3)
Copper	µg/L	5 (3 - 10)	1 (<1 - 11)	3 (2 - 6)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 4)	<1 (<1 - 1)
Zinc	µg/L	20 (15 - 31)	22 (15 - 69)	22 (17 - 35)
Flow	L/s	NM	22 (6 - 160)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Fo Tan Nullah and Kwun Yam Shan Stream) in 2017

Parameter	Unit	<u>Fo Tan Nullah</u>		<u>Kwun Yam Shan Stream</u>
		TR17	TR17L	KY1
Dissolved oxygen	mg/L	9.1 (8.0 - 10.5)	6.9 (5.7 - 8.5)	8.2 (7.9 - 9.8)
pH		8.8 (8.1 - 9.4)	8.0 (7.5 - 8.8)	7.7 (6.8 - 8.5)
Suspended solids	mg/L	4.9 (1.2 - 210.0)	5.5 (1.8 - 26.0)	2.7 (0.9 - 21.0)
5-day Biochemical Oxygen Demand	mg/L	5.9 (2.0 - 260.0)	3.4 (1.6 - 16.0)	0.5 (<0.1 - 1.5)
Chemical Oxygen Demand	mg/L	11 (5 - 350)	12 (6 - 39)	4 (<2 - 10)
Oil & grease	mg/L	<0.5 (<0.5 - 38.0)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	15 000 (700 - 5 500 000)	2 200 (40 - 83 000)	200 (19 - 4 200)
Faecal coliforms	counts/ 100mL	91 000 (6 700 - 6 100 000)	18 000 (490 - 400 000)	1 300 (110 - 21 000)
Ammonia-nitrogen	mg/L	0.125 (0.056 - 4.200)	0.215 (0.059 - 1.900)	0.012 (<0.005 - 0.025)
Nitrate-nitrogen	mg/L	0.625 (0.002 - 1.000)	0.395 (0.054 - 0.970)	0.520 (0.380 - 1.000)
Total Kjeldahl nitrogen	mg/L	0.61 (0.46 - 18.00)	0.38 (0.30 - 2.70)	0.19 (0.07 - 0.47)
Ortho-phosphate Phosphorus	mg/L	0.016 (0.005 - 0.520)	0.009 (<0.002 - 0.110)	0.067 (0.034 - 0.130)
Total phosphorus	mg/L	0.08 (0.03 - 2.90)	0.06 (0.04 - 0.16)	0.08 (0.05 - 0.20)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	135 (83 - 1 400)	81 (<50 - 450)	52 (<50 - 134)
Cadmium	µg/L	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 4)	1 (<1 - 5)	<1 (<1 - <1)
Copper	µg/L	6 (3 - 17)	5 (3 - 16)	<1 (<1 - 3)
Lead	µg/L	1 (<1 - 26)	<1 (<1 - 5)	<1 (<1 - 3)
Zinc	µg/L	34 (21 - 215)	22 (16 - 93)	10 (<10 - 22)
Flow	L/s	51 (35 - 600)	NM	13 (3 - 57)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Tai Wai Nullah and Tin Sum Nullah) in 2017

Parameter	Unit	Tai Wai Nullah		Tin Sum Nullah	
		TR19A	TR19C	TR19	TR20B
Dissolved oxygen	mg/L	8.8 (7.9 - 10.2)	9.2 (8.1 - 10.2)	9.9 (8.3 - 13.7)	8.4 (7.7 - 9.5)
pH		7.8 (7.4 - 8.9)	7.5 (7.0 - 8.3)	7.5 (7.0 - 8.6)	7.6 (7.1 - 10.9)
Suspended solids	mg/L	2.8 (1.3 - 11.0)	2.7 (1.3 - 10.0)	3.0 (0.9 - 9.0)	4.1 (0.7 - 66.0)
5-day Biochemical Oxygen Demand	mg/L	0.8 (0.5 - 20.0)	1.7 (0.5 - 10.0)	1.7 (<0.1 - 9.4)	0.3 (0.1 - 0.8)
Chemical Oxygen Demand	mg/L	4 (<2 - 32)	4 (<2 - 29)	7 (2 - 30)	2 (<2 - 6)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	620 (<1 - 130 000)	6 700 (430 - 690 000)	14 000 (1 500 - 200 000)	2 (<1 - 340)
Faecal coliforms	counts/ 100mL	12 000 (<1 - 1 400 000)	39 000 (2 400 - 700 000)	99 000 (28 000 - 540 000)	4 (<1 - 2 500)
Ammonia-nitrogen	mg/L	0.035 (0.005 - 0.750)	0.050 (0.018 - 1.000)	0.070 (0.016 - 1.000)	0.060 (<0.005 - 0.320)
Nitrate-nitrogen	mg/L	0.945 (0.670 - 2.400)	0.720 (0.510 - 2.000)	0.670 (0.530 - 2.100)	0.990 (0.610 - 1.900)
Total Kjeldahl nitrogen	mg/L	0.31 (0.18 - 2.40)	0.32 (0.20 - 2.70)	0.36 (0.15 - 2.60)	0.27 (0.10 - 0.86)
Ortho-phosphate Phosphorus	mg/L	0.009 (0.005 - 0.036)	0.018 (<0.002 - 0.057)	0.013 (<0.002 - 0.061)	0.012 (0.004 - 0.320)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.10)	0.04 (<0.02 - 0.13)	0.03 (<0.02 - 0.12)	<0.02 (<0.02 - 0.41)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	87 (54 - 225)	99 (66 - 280)	85 (65 - 233)	161 (<50 - 991)
Cadmium	µg/L	0.8 (<0.1 - 4.0)	0.2 (<0.1 - 0.2)	0.2 (<0.1 - 0.5)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	<1 (<1 - 3)	<1 (<1 - 1)
Copper	µg/L	2 (<1 - 18)	5 (2 - 21)	4 (3 - 23)	2 (1 - 6)
Lead	µg/L	1 (<1 - 10)	<1 (<1 - 5)	<1 (<1 - 5)	<1 (<1 - 5)
Zinc	µg/L	58 (24 - 191)	30 (26 - 84)	28 (19 - 86)	18 (11 - 197)
Flow	L/s	27 (8 - 75)	68 (26 - 180)	83 (17 - 240)	26 (7 - 150)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2017 (Part 1 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12H	TR12D	TR12C
Dissolved oxygen	mg/L	8.4 (7.9 - 9.9)	8.5 (7.9 - 10.0)	8.2 (5.8 - 9.6)
pH		7.2 (6.7 - 7.7)	7.1 (6.9 - 8.0)	7.3 (7.0 - 7.6)
Suspended solids	mg/L	1.4 (0.6 - 28.0)	0.7 (<0.5 - 1.5)	2.7 (0.9 - 7.6)
5-day Biochemical Oxygen Demand	mg/L	0.5 (0.1 - 3.8)	0.4 (0.1 - 2.0)	2.9 (1.2 - 6.7)
Chemical Oxygen Demand	mg/L	<2 (<2 - 12)	<2 (<2 - 6)	6 (4 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 200 (350 - 22 000)	370 (53 - 16 000)	13 000 (2 800 - 86 000)
Faecal coliforms	counts/ 100mL	6 800 (930 - 110 000)	3 200 (620 - 38 000)	49 000 (16 000 - 200 000)
Ammonia-nitrogen	mg/L	0.079 (0.024 - 0.260)	0.017 (0.008 - 0.026)	0.415 (0.150 - 0.910)
Nitrate-nitrogen	mg/L	0.645 (0.460 - 1.100)	0.275 (0.059 - 0.800)	1.000 (0.720 - 1.900)
Total Kjeldahl Nitrogen	mg/L	0.23 (0.14 - 0.64)	0.14 (<0.05 - 0.33)	0.76 (0.42 - 1.50)
Ortho-phosphate Phosphorus	mg/L	0.033 (0.005 - 0.092)	0.011 (0.003 - 0.035)	0.115 (0.013 - 0.160)
Total phosphorus	mg/L	0.04 (0.03 - 0.11)	<0.02 (<0.02 - 0.05)	0.16 (0.10 - 0.23)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 215)	<50 (<50 - 56)	<50 (<50 - 110)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 5)	1 (<1 - 4)	1 (<1 - 5)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 1)	<1 (<1 - 1)
Zinc	µg/L	17 (11 - 37)	23 (17 - 43)	15 (12 - 31)
Flow	L/s	54 (5 - 160)	18 (9 - 60)	108 (40 - 150)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2017 (Part 2 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12G	TR12F	TR12B
Dissolved oxygen	mg/L	8.0 (7.6 - 9.7)	8.4 (7.8 - 9.9)	9.1 (8.0 - 10.1)
pH		7.1 (6.7 - 7.6)	7.4 (6.9 - 7.7)	7.4 (7.0 - 8.2)
Suspended solids	mg/L	1.8 (0.8 - 10.0)	1.7 (0.7 - 60.0)	1.2 (0.6 - 6.0)
5-day Biochemical Oxygen Demand	mg/L	0.5 (<0.1 - 1.3)	0.5 (0.2 - 6.5)	0.7 (0.3 - 2.8)
Chemical Oxygen Demand	mg/L	3 (<2 - 8)	3 (<2 - 10)	3 (<2 - 8)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	220 (41 - 5 600)	1 100 (180 - 100 000)	930 (260 - 45 000)
Faecal coliforms	counts/ 100mL	2 900 (340 - 21 000)	6 500 (1 300 - 180 000)	4 100 (870 - 110 000)
Ammonia-nitrogen	mg/L	0.015 (<0.005 - 0.024)	0.026 (0.009 - 0.260)	0.043 (0.026 - 0.074)
Nitrate-nitrogen	mg/L	0.076 (<0.002 - 0.190)	0.410 (0.120 - 0.510)	0.570 (0.280 - 1.100)
Total Kjeldahl nitrogen	mg/L	0.14 (<0.05 - 0.37)	0.17 (0.08 - 0.95)	0.22 (0.12 - 0.62)
Ortho-phosphate Phosphorus	mg/L	0.016 (0.005 - 0.038)	0.020 (0.004 - 0.030)	0.030 (0.005 - 0.050)
Total phosphorus	mg/L	0.03 (<0.02 - 0.17)	0.05 (0.02 - 0.27)	0.04 (0.03 - 0.14)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 51)	<50 (<50 - 377)	<50 (<50 - 108)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 3)	<1 (<1 - 11)	1 (<1 - 5)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 8)	<1 (<1 - 2)
Zinc	µg/L	13 (<10 - 54)	17 (<10 - 46)	16 (13 - 32)
Flow	L/s	18 (8 - 40)	30 (17 - 80)	152 (45 - 390)

- Notes:
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 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River (Part 3 of 3) and Tai Po River in 2017

Parameter	Unit	Lam Tsuen River			Tai Po River
		TR12E	TR12	TR12I	TR13
Dissolved oxygen	mg/L	8.3 (8.0 - 9.6)	8.6 (7.1 - 9.6)	6.1 (3.8 - 8.1)	7.3 (4.9 - 9.0)
pH		8.0 (7.7 - 8.2)	7.4 (7.1 - 7.9)	7.4 (7.1 - 7.6)	7.5 (7.2 - 8.0)
Suspended solids	mg/L	4.7 (1.2 - 670.0)	7.3 (1.6 - 55.0)	5.8 (1.4 - 13.0)	3.6 (1.2 - 43.0)
5-day Biochemical Oxygen Demand	mg/L	0.8 (0.2 - 7.7)	6.0 (1.5 - 11.0)	2.7 (1.4 - 15.0)	0.8 (0.4 - 2.2)
Chemical Oxygen Demand	mg/L	3 (<2 - 41)	11 (5 - 23)	10 (7 - 75)	10 (6 - 16)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	2 200 (570 - 20 000)	2 500 (550 - 9 500)	100 000 (16 000 - 3 500 000)	9 200 (1 500 - 80 000)
Faecal coliforms	counts/ 100mL	7 200 (2 100 - 58 000)	18 000 (1 600 - 1 300 000)	350 000 (78 000 - 7 100 000)	26 000 (3 800 - 250 000)
Ammonia-nitrogen	mg/L	0.078 (0.017 - 0.190)	1.150 (0.130 - 3.000)	0.720 (0.270 - 1.800)	0.230 (0.073 - 0.440)
Nitrate-nitrogen	mg/L	0.715 (0.480 - 1.000)	1.250 (0.760 - 3.200)	0.575 (<0.002 - 1.100)	0.555 (0.190 - 0.810)
Total Kjeldahl nitrogen	mg/L	0.27 (0.09 - 3.70)	1.90 (0.33 - 4.50)	1.06 (0.51 - 3.00)	0.38 (0.16 - 0.86)
Ortho-phosphate Phosphorus	mg/L	0.014 (<0.002 - 0.024)	0.310 (0.075 - 0.610)	0.045 (<0.002 - 0.110)	0.036 (0.014 - 0.095)
Total phosphorus	mg/L	0.04 (<0.02 - 2.10)	0.39 (0.15 - 0.94)	0.13 (0.08 - 0.41)	0.08 (0.04 - 0.19)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	427 (100 - 57 848)	62 (<50 - 228)	74 (51 - 669)	53 (<50 - 346)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - 21)	<1 (<1 - <1)	2 (<1 - 2)	1 (<1 - 3)
Copper	µg/L	2 (<1 - 41)	2 (1 - 8)	3 (2 - 6)	3 (1 - 6)
Lead	µg/L	<1 (<1 - 18)	<1 (<1 - 8)	<1 (<1 - <1)	<1 (<1 - 7)
Zinc	µg/L	18 (<10 - 62)	17 (14 - 37)	18 (15 - 31)	18 (11 - 35)
Flow	L/s	167 (95 - 300)	37 (23 - 90)	NM	113 (73 - 175)

- Notes:
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 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream in 2017

Parameter	Unit	<u>Tai Po Kau Stream</u>	<u>Shan Liu Stream</u>	<u>Tung Tze Stream</u>
		TR14	TR4	TR6
Dissolved oxygen	mg/L	8.2 (7.3 - 9.8)	8.1 (7.6 - 9.6)	6.0 (4.6 - 7.9)
pH		7.4 (6.6 - 7.6)	7.9 (7.1 - 8.1)	7.5 (7.1 - 8.1)
Suspended solids	mg/L	2.4 (<0.5 - 14.0)	2.6 (1.3 - 210.0)	5.6 (2.7 - 27.0)
5-day Biochemical Oxygen Demand	mg/L	0.4 (0.1 - 1.6)	0.9 (0.1 - 2.4)	1.6 (0.6 - 3.3)
Chemical Oxygen Demand	mg/L	6 (<2 - 17)	3 (2 - 12)	15 (10 - 20)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)
<i>E. coli</i>	counts/ 100mL	570 (69 - 27 000)	2 300 (670 - 9 000)	6 700 (1 600 - 130 000)
Faecal coliforms	counts/ 100mL	2 100 (230 - 96 000)	9 000 (1 500 - 54 000)	13 000 (2 500 - 310 000)
Ammonia-nitrogen	mg/L	0.077 (0.019 - 0.170)	0.087 (0.034 - 0.310)	0.465 (0.120 - 1.700)
Nitrate-nitrogen	mg/L	0.250 (0.150 - 0.620)	0.495 (0.170 - 1.400)	0.190 (0.043 - 0.870)
Total Kjeldahl nitrogen	mg/L	0.19 (0.11 - 0.49)	0.33 (0.10 - 0.70)	0.69 (0.24 - 1.90)
Ortho-phosphate Phosphorus	mg/L	0.010 (0.002 - 0.036)	0.046 (0.006 - 0.085)	0.032 (0.004 - 0.150)
Total phosphorus	mg/L	0.03 (<0.02 - 0.07)	0.08 (0.04 - 0.13)	0.10 (0.06 - 0.21)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	52 (<50 - 195)	57 (<50 - 659)	50 (<50 - 148)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - 1)	2 (<1 - 3)
Copper	µg/L	2 (<1 - 8)	1 (<1 - 3)	3 (2 - 7)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 13)	<1 (<1 - 3)
Zinc	µg/L	15 (<10 - 27)	13 (<10 - 22)	15 (<10 - 24)
Flow	L/s	103 (21 - 240)	60 (20 - 135)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ho Chung River in 2017

Parameter	Unit	<u>Ho Chung River</u>	
		PR1	PR2
Dissolved oxygen	mg/L	7.0 (5.2 - 8.4)	8.5 (7.5 - 9.3)
pH		7.3 (6.9 - 7.7)	7.3 (6.9 - 7.5)
Suspended solids	mg/L	4.0 (2.2 - 18.0)	3.1 (2.1 - 6.9)
5-day Biochemical Oxygen Demand	mg/L	1.9 (0.5 - 5.4)	0.7 (0.3 - 2.3)
Chemical Oxygen Demand	mg/L	9 (5 - 14)	2 (<2 - 7)
Oil & grease	mg/L	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - 0.7)
<i>E. coli</i>	counts/ 100mL	13 000 (860 - 92 000)	1 200 (430 - 14 000)
Faecal coliforms	counts/ 100mL	44 000 (4 500 - 280 000)	5 200 (1 100 - 46 000)
Ammonia-nitrogen	mg/L	1.300 (0.044 - 2.100)	0.055 (0.010 - 0.160)
Nitrate-nitrogen	mg/L	0.345 (0.130 - 0.620)	0.275 (0.120 - 0.570)
Total Kjeldahl nitrogen	mg/L	1.80 (0.46 - 2.50)	0.19 (0.11 - 0.39)
Ortho-phosphate Phosphorus	mg/L	0.099 (0.005 - 0.180)	0.015 (0.003 - 0.030)
Total phosphorus	mg/L	0.18 (0.07 - 0.32)	0.03 (<0.02 - 0.07)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	66 (<50 - 160)	57 (<50 - 135)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - <1)
Copper	µg/L	3 (1 - 6)	<1 (<1 - 3)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - <1)
Zinc	µg/L	18 (12 - 32)	12 (<10 - 20)
Flow	L/s	NM	300 (210 - 900)

- Notes:
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 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sha Kok Mei Stream in 2017

Parameter	Unit	Sha Kok Mei Stream	
		PR5	PR6
Dissolved oxygen	mg/L	7.7 (6.8 - 9.0)	8.5 (7.8 - 9.5)
pH		7.4 (7.0 - 8.2)	7.3 (6.5 - 7.7)
Suspended solids	mg/L	2.1 (0.6 - 7.2)	1.4 (0.8 - 5.0)
5-day Biochemical Oxygen Demand	mg/L	1.4 (<0.1 - 25.0)	1.3 (0.3 - 5.2)
Chemical Oxygen Demand	mg/L	4 (3 - 22)	4 (2 - 9)
Oil & grease	mg/L	<0.5 (<0.5 - 1.6)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	230 (<10 - 13 000)	15 000 (3 300 - 44 000)
Faecal coliforms	counts/ 100mL	680 (10 - 41 000)	33 000 (9 400 - 110 000)
Ammonia-nitrogen	mg/L	0.150 (0.015 - 0.290)	0.155 (0.033 - 0.380)
Nitrate-nitrogen	mg/L	0.646 (<0.002 - 1.100)	1.500 (1.100 - 2.400)
Total Kjeldahl nitrogen	mg/L	0.39 (0.21 - 1.50)	0.50 (0.24 - 1.00)
Ortho-phosphate Phosphorus	mg/L	0.039 (0.003 - 0.061)	0.075 (0.018 - 0.110)
Total phosphorus	mg/L	0.08 (<0.02 - 0.22)	0.10 (0.06 - 0.13)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	57 (<50 - 89)	54 (<50 - 114)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 5)	2 (1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 2)
Zinc	µg/L	13 (<10 - 22)	13 (10 - 28)
Flow	L/s	95 (6 - 300)	NM

- Notes:
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 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Chung Hau Stream in 2017

Parameter	Unit	<u>Tai Chung Hau Stream</u>	
		PR7	PR8
Dissolved oxygen	mg/L	8.6 (7.5 - 9.8)	8.1 (7.4 - 9.0)
pH		7.8 (6.9 - 8.5)	7.8 (7.0 - 10.7)
Suspended solids	mg/L	2.7 (1.7 - 6.3)	2.7 (1.4 - 5.8)
5-day Biochemical Oxygen Demand	mg/L	1.1 (0.6 - 2.4)	4.3 (0.8 - 7.6)
Chemical Oxygen Demand	mg/L	4 (2 - 9)	6 (2 - 20)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	6 400 (2 500 - 44 000)	13 000 (2 800 - 50 000)
Faecal coliforms	counts/ 100mL	24 000 (7 500 - 200 000)	35 000 (8 400 - 230 000)
Ammonia-nitrogen	mg/L	0.110 (0.042 - 0.190)	0.069 (0.026 - 0.330)
Nitrate-nitrogen	mg/L	0.730 (0.330 - 1.200)	0.985 (0.590 - 1.500)
Total Kjeldahl Nitrogen	mg/L	0.34 (0.18 - 0.65)	0.57 (0.19 - 1.60)
Ortho-phosphate Phosphorus	mg/L	0.047 (0.007 - 0.082)	0.068 (0.009 - 0.130)
Total phosphorus	mg/L	0.08 (0.04 - 0.11)	0.09 (0.05 - 0.15)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	81 (<50 - 181)	68 (<50 - 275)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - 13)
Copper	µg/L	2 (<1 - 7)	2 (<1 - 7)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 1)
Zinc	µg/L	16 (<10 - 36)	14 (<10 - 31)
Flow	L/s	120 (30 - 300)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tseng Lan Shue Stream in 2017

Parameter	Unit	Tseng Lan Shue Stream		
		JR3	JR6	JR11
Dissolved oxygen	mg/L	6.3 (5.0 - 8.0)	8.0 (7.2 - 8.6)	8.6 (8.0 - 9.7)
pH		7.1 (6.5 - 7.3)	7.4 (6.9 - 7.7)	7.5 (6.5 - 7.8)
Suspended solids	mg/L	3.9 (1.5 - 37.0)	5.5 (1.4 - 16.0)	3.0 (1.4 - 16.0)
5-day Biochemical Oxygen Demand	mg/L	9.0 (1.8 - 29.0)	4.8 (1.8 - 14.0)	1.9 (0.7 - 4.3)
Chemical Oxygen Demand	mg/L	19 (7 - 42)	10 (6 - 24)	7 (3 - 12)
Oil & grease	mg/L	<0.5 (<0.5 - 1.1)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.5)
<i>E. coli</i>	counts/ 100mL	70 000 (37 000 - 150 000)	41 000 (14 000 - 480 000)	1 200 (210 - 31 000)
Faecal coliforms	counts/ 100mL	130 000 (76 000 - 370 000)	78 000 (32 000 - 590 000)	4 800 (1 000 - 75 000)
Ammonia-nitrogen	mg/L	5.200 (0.530 - 9.600)	0.290 (0.080 - 0.660)	0.077 (0.022 - 0.210)
Nitrate-nitrogen	mg/L	1.300 (0.480 - 1.700)	2.000 (1.300 - 2.700)	2.850 (0.770 - 4.400)
Total Kjeldahl nitrogen	mg/L	5.80 (0.84 - 11.00)	1.01 (0.78 - 2.10)	0.54 (0.27 - 1.00)
Ortho-phosphate Phosphorus	mg/L	0.510 (0.120 - 0.910)	0.430 (0.140 - 0.700)	0.395 (0.070 - 0.580)
Total phosphorus	mg/L	0.67 (0.17 - 1.10)	0.55 (0.21 - 0.95)	0.45 (0.14 - 0.63)
Total sulphide	mg/L	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	99 (<50 - 409)	86 (56 - 253)	52 (<50 - 146)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (2 - 7)	3 (1 - 10)	2 (1 - 5)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 5)	<1 (<1 - 2)
Zinc	µg/L	26 (20 - 71)	26 (18 - 70)	20 (14 - 31)
Flow	L/s	NM	NM	80 (44 - 960)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Indus in 2017

Parameter	Unit	River Indus		
		IN1	IN2	IN3
Dissolved oxygen	mg/L	5.1 (2.4 - 7.4)	6.9 (5.8 - 8.7)	7.6 (6.4 - 9.4)
pH		7.1 (6.8 - 7.4)	7.4 (6.9 - 7.6)	9.4 (7.3 - 10.5)
Suspended solids	mg/L	17.0 (4.6 - 360.0)	6.0 (2.0 - 390.0)	5.1 (2.0 - 280.0)
5-day Biochemical Oxygen Demand	mg/L	5.1 (2.4 - 57.0)	2.2 (0.9 - 21.0)	2.1 (0.7 - 24.0)
Chemical Oxygen Demand	mg/L	25 (8 - 84)	7 (4 - 51)	8 (5 - 41)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	51 000 (2 800 - 2 400 000)	2 600 (380 - 120 000)	810 (20 - 55 000)
Faecal coliforms	counts/ 100mL	140 000 (11 000 - 4 300 000)	15 000 (2 700 - 340 000)	5 300 (120 - 210 000)
Ammonia-nitrogen	mg/L	1.500 (0.400 - 9.800)	0.580 (0.240 - 1.400)	0.155 (0.021 - 0.350)
Nitrate-nitrogen	mg/L	2.400 (<0.002 - 5.900)	1.150 (0.730 - 1.400)	0.820 (0.500 - 1.900)
Total Kjeldahl nitrogen	mg/L	3.30 (1.10 - 13.00)	0.99 (0.45 - 3.30)	0.64 (0.32 - 1.50)
Ortho-phosphate Phosphorus	mg/L	0.205 (0.042 - 0.490)	0.076 (0.032 - 0.120)	0.052 (0.004 - 0.470)
Total phosphorus	mg/L	0.48 (0.15 - 1.40)	0.14 (0.10 - 1.00)	0.08 (0.06 - 0.60)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	171 (70 - 1 449)	124 (82 - 1 874)	396 (153 - 1 042)
Cadmium	µg/L	<0.1 (<0.1 - 0.4)	<0.1 (<0.1 - 0.4)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	1 (<1 - 6)	<1 (<1 - 4)	5 (<1 - 13)
Copper	µg/L	4 (2 - 30)	2 (1 - 29)	2 (2 - 6)
Lead	µg/L	2 (<1 - 28)	1 (<1 - 48)	<1 (<1 - 6)
Zinc	µg/L	30 (17 - 195)	18 (15 - 220)	18 (<10 - 41)
Flow	L/s	NM	NM	45 (15 - 440)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Beas in 2017

Parameter	Unit	River Beas		
		RB1	RB2	RB3
Dissolved oxygen	mg/L	9.1 (8.0 - 10.4)	8.0 (6.1 - 9.3)	7.8 (5.5 - 10.2)
pH		7.7 (7.1 - 8.1)	7.3 (6.8 - 7.6)	7.4 (6.9 - 7.6)
Suspended solids	mg/L	3.9 (2.0 - 210.0)	5.0 (3.2 - 42.0)	9.8 (4.6 - 110.0)
5-day Biochemical Oxygen Demand	mg/L	1.9 (1.0 - 19.0)	4.6 (1.2 - 37.0)	6.4 (1.5 - 28.0)
Chemical Oxygen Demand	mg/L	6 (4 - 51)	10 (6 - 27)	14 (8 - 34)
Oil & grease	mg/L	<0.5 (<0.5 - 3.7)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 000 (1 200 - 76 000)	2 600 (450 - 41 000)	14 000 (1 500 - 150 000)
Faecal coliforms	counts/ 100mL	17 000 (1 900 - 430 000)	14 000 (2 900 - 90 000)	54 000 (9 900 - 440 000)
Ammonia-nitrogen	mg/L	0.210 (0.053 - 0.480)	1.070 (0.200 - 1.800)	1.600 (0.140 - 2.500)
Nitrate-nitrogen	mg/L	1.050 (0.450 - 1.500)	0.640 (0.280 - 1.000)	0.785 (0.200 - 2.400)
Total Kjeldahl nitrogen	mg/L	0.64 (0.26 - 1.70)	1.50 (0.48 - 3.20)	2.25 (0.48 - 4.00)
Ortho-phosphate Phosphorus	mg/L	0.215 (0.088 - 0.360)	0.140 (0.078 - 0.280)	0.175 (0.028 - 0.570)
Total phosphorus	mg/L	0.28 (0.14 - 1.10)	0.27 (0.17 - 0.83)	0.34 (0.18 - 0.88)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	51 (<50 - 858)	61 (<50 - 219)	90 (<50 - 778)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 3)
Copper	µg/L	2 (1 - 17)	2 (2 - 7)	4 (2 - 29)
Lead	µg/L	<1 (<1 - 18)	<1 (<1 - 3)	1 (<1 - 14)
Zinc	µg/L	16 (<10 - 146)	21 (13 - 53)	43 (17 - 239)
Flow	L/s	175 (50 - 1 575)	164 (60 - 6 480)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Ganges in 2017

Parameter	Unit	River Ganges		
		GR1	GR2	GR3
Dissolved oxygen	mg/L	7.4 (4.4 - 9.0)	6.5 (6.0 - 8.4)	8.1 (6.9 - 8.9)
pH		7.4 (7.0 - 7.5)	7.1 (6.6 - 7.3)	7.4 (6.6 - 7.8)
Suspended solids	mg/L	8.4 (3.8 - 190.0)	12.0 (5.2 - 170.0)	2.6 (1.0 - 100.0)
5-day Biochemical Oxygen Demand	mg/L	3.4 (1.2 - 29.0)	2.6 (1.0 - 11.0)	0.4 (<0.1 - 12.0)
Chemical Oxygen Demand	mg/L	12 (7 - 59)	12 (7 - 34)	4 (<2 - 29)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	12 000 (1 700 - 480 000)	11 000 (1 700 - 210 000)	1 000 (61 - 90 000)
Faecal coliforms	counts/ 100mL	30 000 (4 200 - 830 000)	29 000 (3 700 - 710 000)	6 700 (270 - 380 000)
Ammonia-nitrogen	mg/L	1.900 (0.270 - 7.100)	0.910 (0.160 - 13.000)	0.088 (0.017 - 0.300)
Nitrate-nitrogen	mg/L	1.250 (0.670 - 3.000)	1.350 (0.600 - 3.800)	0.375 (0.140 - 0.920)
Total Kjeldahl nitrogen	mg/L	2.15 (0.71 - 7.70)	2.05 (0.42 - 14.00)	0.33 (0.11 - 0.79)
Ortho-phosphate Phosphorus	mg/L	0.420 (0.250 - 1.600)	0.160 (0.080 - 0.540)	0.003 (<0.002 - 0.006)
Total phosphorus	mg/L	0.58 (0.36 - 1.90)	0.39 (0.20 - 0.85)	<0.02 (<0.02 - 0.05)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	97 (51 - 922)	79 (<50 - 791)	<50 (<50 - 690)
Cadmium	µg/L	<0.1 (<0.1 - 0.4)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.3)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 2)
Copper	µg/L	3 (2 - 47)	3 (2 - 16)	2 (<1 - 11)
Lead	µg/L	1 (<1 - 19)	2 (<1 - 12)	<1 (<1 - 7)
Zinc	µg/L	22 (14 - 205)	27 (16 - 83)	18 (11 - 62)
Flow	L/s	38 (15 - 473)	34 (20 - 320)	75 (30 - 300)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2017 (Part 1 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL1	YL2
Dissolved oxygen	mg/L	6.0 (3.6 - 7.8)	6.0 (3.5 - 7.8)
pH		7.3 (6.7 - 7.5)	7.4 (6.9 - 7.6)
Suspended solids	mg/L	7.4 (3.6 - 35.0)	6.3 (2.5 - 12.0)
5-day Biochemical Oxygen Demand	mg/L	11.3 (3.7 - 59.0)	6.9 (2.7 - 24.0)
Chemical Oxygen Demand	mg/L	24 (6 - 57)	33 (16 - 61)
Oil & grease	mg/L	<0.5 (<0.5 - 1.4)	0.7 (<0.5 - 2.4)
<i>E. coli</i>	counts/ 100mL	81 000 (900 - 390 000)	72 000 (7 000 - 730 000)
Faecal coliforms	counts/ 100mL	210 000 (3 300 - 3 600 000)	130 000 (17 000 - 780 000)
Ammonia-nitrogen	mg/L	5.750 (0.570 - 17.000)	17.500 (4.000 - 35.000)
Nitrate-nitrogen	mg/L	0.510 (0.006 - 1.100)	0.830 (0.046 - 2.800)
Total Kjeldahl nitrogen	mg/L	6.35 (1.20 - 19.00)	18.50 (4.90 - 35.00)
Ortho-phosphate Phosphorus	mg/L	1.050 (0.006 - 1.800)	2.600 (0.660 - 3.400)
Total phosphorus	mg/L	1.20 (0.22 - 2.00)	2.95 (0.76 - 3.60)
Total sulphide	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - 0.22)
Aluminium	µg/L	108 (81 - 670)	76 (<50 - 373)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)
Copper	µg/L	5 (2 - 12)	3 (2 - 5)
Lead	µg/L	1 (<1 - 11)	<1 (<1 - 3)
Zinc	µg/L	29 (22 - 60)	28 (20 - 58)
Flow	L/s	230 (110 - 450)	23 (9 - 44)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2017 (Part 2 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL3	YL4
Dissolved oxygen	mg/L	4.3 (2.5 - 6.3)	3.4 (2.3 - 5.7)
pH		7.4 (6.7 - 7.6)	7.2 (6.7 - 7.8)
Suspended solids	mg/L	20.0 (11.0 - 57.0)	37.0 (8.2 - 57.0)
5-day Biochemical Oxygen Demand	mg/L	35.0 (2.0 - 80.0)	110.0 (2.0 - 200.0)
Chemical Oxygen Demand	mg/L	41 (13 - 97)	98 (18 - 220)
Oil & grease	mg/L	1.0 (<0.5 - 11.0)	2.4 (<0.5 - 34.0)
<i>E. coli</i>	counts/ 100mL	830 000 (200 000 - 4 200 000)	2 300 000 (260 000 - 31 000 000)
Faecal coliforms	counts/ 100mL	1 800 000 (310 000 - 7 600 000)	5 300 000 (460 000 - 48 000 000)
Ammonia-nitrogen	mg/L	8.100 (1.900 - 29.000)	7.350 (2.000 - 10.000)
Nitrate-nitrogen	mg/L	0.006 (<0.002 - 0.990)	<0.002 (<0.002 - 0.820)
Total Kjeldahl nitrogen	mg/L	9.85 (2.60 - 32.00)	11.50 (2.80 - 17.00)
Ortho-phosphate Phosphorus	mg/L	0.730 (0.220 - 2.400)	0.395 (0.041 - 0.940)
Total phosphorus	mg/L	1.30 (0.42 - 3.20)	1.25 (0.31 - 1.60)
Total sulphide	mg/L	0.04 (<0.02 - 0.12)	0.17 (<0.02 - 0.52)
Aluminium	µg/L	145 (87 - 303)	167 (102 - 359)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 3)
Copper	µg/L	6 (3 - 16)	6 (5 - 14)
Lead	µg/L	2 (1 - 7)	3 (1 - 10)
Zinc	µg/L	39 (21 - 73)	44 (28 - 91)
Flow	L/s	530 (293 - 1 460)	165 (90 - 213)

- Notes:
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 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kam Tin River in 2017

Parameter	Unit	Kam Tin River	
		KT1	KT2
Dissolved oxygen	mg/L	5.4 (3.4 - 7.3)	5.1 (2.7 - 7.5)
pH		7.3 (6.7 - 7.5)	7.5 (6.9 - 7.7)
Suspended solids	mg/L	8.5 (5.2 - 34.0)	17.0 (4.6 - 74.0)
5-day Biochemical Oxygen Demand	mg/L	15.5 (4.2 - 53.0)	18.0 (4.9 - 170.0)
Chemical Oxygen Demand	mg/L	19 (7 - 70)	28 (11 - 180)
Oil & grease	mg/L	<0.5 (<0.5 - 1.3)	<0.5 (<0.5 - 4.2)
<i>E. coli</i>	counts/ 100mL	160 000 (16 000 - 1 400 000)	140 000 (13 000 - 590 000)
Faecal coliforms	counts/ 100mL	320 000 (79 000 - 1 900 000)	300 000 (33 000 - 950 000)
Ammonia-nitrogen	mg/L	5.750 (1.000 - 25.000)	6.900 (1.300 - 19.000)
Nitrate-nitrogen	mg/L	0.555 (<0.002 - 1.200)	0.280 (<0.002 - 1.100)
Total Kjeldahl nitrogen	mg/L	6.95 (1.90 - 26.00)	7.95 (2.00 - 31.00)
Ortho-phosphate Phosphorus	mg/L	1.125 (0.390 - 3.200)	1.200 (0.380 - 2.600)
Total phosphorus	mg/L	1.35 (0.51 - 3.90)	1.60 (0.47 - 5.00)
Total sulphide	mg/L	0.02 (<0.02 - 0.13)	0.02 (<0.02 - 0.10)
Aluminium	µg/L	58 (<50 - 205)	96 (51 - 194)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	11 (4 - 45)	6 (2 - 14)
Lead	µg/L	<1 (<1 - 3)	1 (<1 - 3)
Zinc	µg/L	35 (25 - 91)	37 (16 - 56)
Flow	L/s	378 (55 - 880)	334 (105 - 1 440)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tin Shui Wai Nullah and Fairview Park Nullah in 2017

Parameter	Unit	Tin Shui Wai Nullah		Fairview Park Nullah
		TSR1	TSR2	FVR1
Dissolved oxygen	mg/L	7.2 (3.9 - 11.4)	10.3 (8.0 - 10.9)	6.0 (3.9 - 8.0)
pH		7.5 (7.3 - 8.6)	8.4 (7.1 - 9.3)	7.5 (7.3 - 7.7)
Suspended solids	mg/L	6.3 (1.2 - 19.0)	4.2 (0.6 - 29.0)	20.5 (10.0 - 95.0)
5-day Biochemical Oxygen Demand	mg/L	6.6 (3.5 - 11.0)	1.8 (0.2 - 7.2)	4.9 (2.7 - 14.0)
Chemical Oxygen Demand	mg/L	15 (6 - 20)	6 (2 - 17)	24 (9 - 50)
Oil & grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.7)
<i>E. coli</i>	counts/ 100mL	28 000 (11 000 - 90 000)	27 000 (9 000 - 260 000)	21 000 (3 400 - 120 000)
Faecal coliforms	counts/ 100mL	230 000 (83 000 - 720 000)	67 000 (12 000 - 430 000)	56 000 (12 000 - 230 000)
Ammonia-nitrogen	mg/L	2.700 (0.120 - 4.600)	0.635 (0.058 - 2.900)	2.400 (0.640 - 3.800)
Nitrate-nitrogen	mg/L	0.615 (0.230 - 0.800)	0.845 (0.450 - 1.000)	0.725 (0.260 - 1.200)
Total Kjeldahl nitrogen	mg/L	3.40 (0.28 - 5.20)	1.03 (0.13 - 3.50)	3.00 (1.60 - 5.50)
Ortho-phosphate Phosphorus	mg/L	0.190 (0.025 - 0.420)	0.070 (0.014 - 0.300)	0.365 (0.140 - 0.490)
Total phosphorus	mg/L	0.35 (0.06 - 0.56)	0.09 (0.02 - 0.34)	0.60 (0.28 - 0.83)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	105 (<50 - 236)	147 (92 - 323)	185 (98 - 650)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	<1 (<1 - 4)
Copper	µg/L	3 (1 - 6)	2 (<1 - 7)	3 (2 - 10)
Lead	µg/L	1 (<1 - 3)	<1 (<1 - 5)	2 (1 - 6)
Zinc	µg/L	24 (17 - 108)	17 (<10 - 62)	30 (21 - 79)
Flow	L/s	NM	51 (23 - 225)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ha Pak Nai Stream, Pak Nai Stream and Sheung Pak Nai Stream in 2017

Parameter	Unit	Ha Pak Nai Stream	Pak Nai Stream	Sheung Pak Nai Stream
		DB1	DB3	DB5
Dissolved oxygen	mg/L	8.8 (7.8 - 10.7)	8.5 (7.3 - 10.5)	8.3 (7.2 - 10.1)
pH		7.1 (6.5 - 7.9)	6.8 (6.0 - 7.4)	6.9 (6.4 - 7.5)
Suspended solids	mg/L	3.1 (<0.5 - 43.0)	4.1 (1.0 - 23.0)	7.2 (3.3 - 130.0)
5-day Biochemical Oxygen Demand	mg/L	<0.3 (<0.1 - 2.5)	0.3 (<0.1 - 1.5)	1.5 (0.3 - 6.4)
Chemical Oxygen Demand	mg/L	2 (<2 - 20)	<2 (<2 - 9)	4 (<2 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	110 (18 - 8 200)	290 (61 - 4 000)	14 000 (3 100 - 150 000)
Faecal coliforms	counts/ 100mL	620 (83 - 11 000)	1 300 (260 - 5 200)	21 000 (6 300 - 150 000)
Ammonia-nitrogen	mg/L	0.009 (<0.005 - 0.610)	0.013 (0.008 - 0.260)	0.360 (0.084 - 1.600)
Nitrate-nitrogen	mg/L	0.320 (0.230 - 1.200)	0.240 (0.200 - 0.760)	0.260 (0.210 - 0.590)
Total Kjeldahl nitrogen	mg/L	0.11 (<0.05 - 1.10)	0.08 (<0.05 - 0.59)	0.74 (0.13 - 4.50)
Ortho-phosphate Phosphorus	mg/L	0.005 (<0.002 - 0.006)	0.004 (<0.002 - 0.007)	0.027 (0.005 - 0.240)
Total phosphorus	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	0.13 (<0.02 - 0.45)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	53 (<50 - 272)	<50 (<50 - 177)	82 (<50 - 378)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - 7)	<1 (<1 - 4)	1 (<1 - 13)
Zinc	µg/L	11 (<10 - 23)	14 (<10 - 21)	18 (<10 - 29)
Flow	L/s	40 (12 - 80)	40 (12 - 105)	38 (18 - 100)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream in 2017

Parameter	Unit	<u>Ngau Hom Sha Stream</u>	<u>Tai Shui Hang Stream</u>	<u>Tsang Kok Stream</u>
		DB6	DB2	DB8
Dissolved oxygen	mg/L	7.6 (5.8 - 9.4)	8.6 (7.6 - 10.6)	8.8 (7.6 - 10.0)
pH		6.8 (6.4 - 7.0)	7.3 (6.7 - 7.9)	7.5 (6.7 - 7.8)
Suspended solids	mg/L	5.2 (2.0 - 55.0)	2.4 (<0.5 - 42.0)	6.3 (1.8 - 120.0)
5-day Biochemical Oxygen Demand	mg/L	0.6 (<0.1 - 1.8)	0.5 (<0.1 - 1.9)	0.8 (0.1 - 3.5)
Chemical Oxygen Demand	mg/L	4 (<2 - 13)	<2 (<2 - 14)	4 (<2 - 14)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	310 (100 - 9 000)	210 (52 - 3 800)	700 (110 - 15 000)
Faecal coliforms	counts/ 100mL	2 400 (420 - 12 000)	1 400 (310 - 8 400)	5 600 (870 - 26 000)
Ammonia-nitrogen	mg/L	0.120 (0.077 - 0.290)	0.016 (<0.005 - 0.580)	0.050 (0.013 - 2.300)
Nitrate-nitrogen	mg/L	0.410 (0.250 - 0.790)	0.370 (0.180 - 0.750)	1.300 (0.740 - 2.300)
Total Kjeldahl nitrogen	mg/L	0.26 (0.18 - 0.67)	0.14 (<0.05 - 1.10)	0.38 (0.15 - 2.80)
Ortho-phosphate Phosphorus	mg/L	0.070 (0.006 - 0.230)	0.003 (<0.002 - 0.006)	0.006 (0.003 - 0.009)
Total phosphorus	mg/L	0.10 (0.05 - 0.27)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.09)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	64 (<50 - 235)	80 (<50 - 288)	115 (57 - 493)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 12)	<1 (<1 - 2)	1 (<1 - 4)
Lead	µg/L	2 (<1 - 9)	<1 (<1 - 6)	1 (<1 - 17)
Zinc	µg/L	25 (<10 - 78)	14 (<10 - 26)	17 (10 - 33)
Flow	L/s	15 (2 - 150)	200 (120 - 500)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2017 (Part 1 of 2)

Parameter	Unit	Mui Wo River		
		MW1	MW2	MW3
Dissolved oxygen	mg/L	8.2 (6.9 - 9.3)	8.5 (6.3 - 10.3)	8.7 (8.0 - 9.5)
pH		7.6 (7.2 - 8.1)	7.6 (6.9 - 8.1)	7.0 (6.2 - 7.5)
Suspended solids	mg/L	1.9 (0.7 - 6.3)	3.0 (1.4 - 5.4)	0.9 (<0.5 - 4.8)
5-day Biochemical Oxygen Demand	mg/L	0.5 (0.1 - 2.3)	0.9 (0.4 - 2.2)	0.4 (0.2 - 0.8)
Chemical Oxygen Demand	mg/L	5 (2 - 8)	8 (3 - 12)	<2 (<2 - 3)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	790 (320 - 3 400)	1 900 (460 - 18 000)	150 (31 - 2 700)
Faecal coliforms	counts/ 100mL	5 300 (670 - 15 000)	8 300 (1 800 - 31 000)	2 600 (590 - 17 000)
Ammonia-nitrogen	mg/L	0.043 (0.021 - 0.085)	0.225 (0.034 - 0.860)	0.013 (<0.005 - 0.033)
Nitrate-nitrogen	mg/L	0.355 (0.160 - 0.510)	0.255 (0.092 - 0.470)	0.310 (0.170 - 0.590)
Total Kjeldahl nitrogen	mg/L	0.17 (0.11 - 0.30)	0.35 (0.19 - 1.20)	0.15 (<0.05 - 0.28)
Ortho-phosphate Phosphorus	mg/L	0.072 (0.012 - 0.130)	0.040 (<0.002 - 0.130)	0.046 (0.015 - 0.084)
Total phosphorus	mg/L	0.10 (0.05 - 0.20)	0.08 (0.05 - 0.19)	0.06 (0.03 - 0.12)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	59 (<50 - 99)	<50 (<50 - 67)	<50 (<50 - 65)
Cadmium	µg/L	<0.1 (<0.1 - 1.4)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 3)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 7)	1 (<1 - 6)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - <1)
Zinc	µg/L	17 (<10 - 26)	15 (<10 - 32)	16 (11 - 25)
Flow	L/s	60 (12 - 100)	NM	20 (6 - 80)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2017 (Part 2 of 2)

Parameter	Unit	<u>Mui Wo River</u>	
		MW4	MW5
Dissolved oxygen	mg/L	7.3 (6.1 - 8.4)	7.8 (4.6 - 9.1)
pH		7.2 (7.0 - 7.7)	7.4 (7.0 - 7.9)
Suspended solids	mg/L	7.5 (1.0 - 22.0)	7.0 (2.9 - 16.0)
5-day Biochemical Oxygen Demand	mg/L	1.1 (0.2 - 2.4)	1.9 (0.8 - 4.6)
Chemical Oxygen Demand	mg/L	8 (5 - 15)	6 (4 - 14)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	470 (60 - 2 200)	5 600 (810 - 20 000)
Faecal coliforms	counts/ 100mL	1 700 (110 - 8 900)	17 000 (1 600 - 94 000)
Ammonia-nitrogen	mg/L	0.340 (0.060 - 0.930)	0.850 (0.180 - 1.300)
Nitrate-nitrogen	mg/L	0.320 (0.200 - 0.570)	0.220 (0.140 - 0.310)
Total Kjeldahl nitrogen	mg/L	0.47 (0.19 - 1.00)	1.11 (0.29 - 1.60)
Ortho-phosphate Phosphorus	mg/L	0.020 (0.002 - 0.060)	0.050 (0.004 - 0.130)
Total phosphorus	mg/L	0.10 (0.05 - 0.16)	0.14 (0.06 - 0.33)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	77 (<50 - 128)	75 (<50 - 140)
Cadmium	µg/L	<0.1 (<0.1 - 0.5)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 5)	<1 (<1 - 3)
Copper	µg/L	3 (1 - 5)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - 2)	1 (<1 - 3)
Zinc	µg/L	22 (17 - 80)	15 (12 - 23)
Flow	L/s	60 (21 - 300)	160 (40 - 240)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tung Chung River in 2017

Parameter	Unit	Tung Chung River		
		TC1	TC2	TC3
Dissolved oxygen	mg/L	8.1 (6.2 - 8.9)	8.7 (7.5 - 9.8)	8.4 (7.8 - 9.0)
pH		6.8 (6.1 - 7.3)	7.6 (7.0 - 8.7)	7.8 (7.0 - 8.4)
Suspended solids	mg/L	1.0 (<0.5 - 2.6)	1.7 (0.8 - 4.3)	2.4 (<0.5 - 12.0)
5-day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 0.6)	0.7 (0.2 - 1.7)	6.7 (1.2 - 23.0)
Chemical Oxygen Demand	mg/L	2 (<2 - 5)	4 (2 - 7)	8 (3 - 25)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 2.4)
<i>E. coli</i>	counts/ 100mL	98 (18 - 660)	550 (96 - 4 500)	10 000 (310 - 30 000)
Faecal coliforms	counts/ 100mL	1 100 (110 - 4 700)	4 000 (1 500 - 10 000)	53 000 (6 800 - 200 000)
Ammonia-nitrogen	mg/L	0.014 (<0.005 - 0.032)	0.028 (0.008 - 0.150)	0.455 (0.160 - 1.200)
Nitrate-nitrogen	mg/L	0.115 (0.052 - 0.500)	0.068 (<0.002 - 0.200)	0.150 (0.041 - 0.280)
Total Kjeldahl nitrogen	mg/L	0.10 (<0.05 - 0.24)	0.18 (0.06 - 0.41)	0.79 (0.33 - 2.50)
Ortho-phosphate Phosphorus	mg/L	0.005 (<0.002 - 0.011)	0.009 (0.003 - 0.013)	0.025 (0.004 - 0.095)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.02)	0.02 (<0.02 - 0.03)	0.08 (0.03 - 0.20)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - <50)	<50 (<50 - 57)	<50 (<50 - 75)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 4)	2 (<1 - 9)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - <1)
Zinc	µg/L	11 (<10 - 17)	13 (<10 - 59)	15 (<10 - 28)
Flow	L/s	46 (11 - 225)	75 (24 - 320)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2017 (Part 1 of 2)

Parameter	Unit	Tuen Mun River		
		TN1	TN2	TN3
Dissolved oxygen	mg/L	4.4 (3.4 - 7.6)	8.8 (7.3 - 10.7)	5.7 (4.4 - 7.4)
pH		7.7 (7.3 - 7.8)	7.5 (7.0 - 7.9)	7.8 (7.5 - 8.2)
Suspended solids	mg/L	14.5 (4.7 - 32.0)	4.6 (1.4 - 23.0)	5.7 (1.3 - 14.0)
5-day Biochemical Oxygen Demand	mg/L	23.0 (6.0 - 33.0)	4.1 (0.3 - 26.0)	2.3 (1.0 - 15.0)
Chemical Oxygen Demand	mg/L	26 (12 - 39)	9 (3 - 39)	14 (<2 - 31)
Oil & grease	mg/L	0.5 (<0.5 - 2.7)	<0.5 (<0.5 - 1.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	120 000 (70 000 - 220 000)	46 000 (3 300 - 1 100 000)	9 800 (900 - 100 000)
Faecal coliforms	counts/ 100mL	830 000 (110 000 - 3 800 000)	110 000 (18 000 - 1 500 000)	68 000 (7 900 - 990 000)
Ammonia-nitrogen	mg/L	6.700 (2.700 - 8.200)	1.350 (0.200 - 8.500)	0.450 (0.240 - 0.790)
Nitrate-nitrogen	mg/L	0.515 (0.019 - 1.600)	1.600 (0.480 - 3.900)	0.455 (0.160 - 1.100)
Total Kjeldahl nitrogen	mg/L	8.05 (3.00 - 10.00)	2.40 (0.42 - 9.80)	0.71 (0.41 - 1.30)
Ortho-phosphate Phosphorus	mg/L	0.575 (0.360 - 0.770)	0.180 (0.017 - 0.530)	0.031 (0.010 - 0.051)
Total phosphorus	mg/L	0.86 (0.46 - 0.98)	0.29 (0.06 - 0.85)	0.09 (0.05 - 0.32)
Total sulphide	mg/L	0.03 (<0.02 - 0.06)	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	108 (69 - 300)	130 (<50 - 347)	93 (<50 - 155)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	1 (<1 - 3)
Copper	µg/L	4 (3 - 7)	3 (1 - 15)	5 (2 - 8)
Lead	µg/L	1 (<1 - 6)	1 (<1 - 8)	<1 (<1 - 2)
Zinc	µg/L	27 (18 - 51)	22 (13 - 40)	20 (15 - 33)
Flow	L/s	155 (50 - 368)	15 (6 - 455)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2017 (Part 2 of 2)

Parameter	Unit	Tuen Mun River		
		TN4	TN5	TN6
Dissolved oxygen	mg/L	6.0 (5.2 - 7.8)	6.0 (4.9 - 7.5)	5.7 (4.3 - 7.2)
pH		7.7 (7.5 - 8.1)	7.7 (7.6 - 8.2)	7.6 (7.4 - 8.0)
Suspended solids	mg/L	6.3 (1.0 - 14.0)	6.0 (2.0 - 14.0)	5.8 (1.4 - 16.0)
5-day Biochemical Oxygen Demand	mg/L	1.7 (0.6 - 2.6)	1.8 (1.0 - 4.3)	2.2 (0.3 - 3.5)
Chemical Oxygen Demand	mg/L	12 (<2 - 17)	10 (<2 - 16)	11 (2 - 20)
Oil & grease	mg/L	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 400 (100 - 34 000)	3 400 (<100 - 100 000)	6 100 (260 - 120 000)
Faecal coliforms	counts/ 100mL	23 000 (300 - 200 000)	22 000 (600 - 770 000)	32 000 (630 - 950 000)
Ammonia-nitrogen	mg/L	0.500 (0.270 - 0.660)	0.420 (0.280 - 0.740)	0.450 (0.240 - 0.690)
Nitrate-nitrogen	mg/L	0.395 (0.180 - 1.100)	0.435 (0.170 - 0.880)	0.430 (0.160 - 0.920)
Total Kjeldahl nitrogen	mg/L	0.71 (0.40 - 0.92)	0.59 (0.36 - 1.10)	0.62 (0.45 - 1.00)
Ortho-phosphate Phosphorus	mg/L	0.043 (0.023 - 0.110)	0.039 (0.019 - 0.062)	0.036 (<0.002 - 0.079)
Total phosphorus	mg/L	0.10 (0.06 - 0.14)	0.09 (0.06 - 0.14)	0.09 (0.05 - 0.14)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	92 (<50 - 178)	83 (<50 - 227)	58 (<50 - 158)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 3)	1 (<1 - 3)	1 (<1 - 5)
Copper	µg/L	5 (2 - 8)	5 (2 - 6)	5 (2 - 10)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 2)	<1 (<1 - 2)
Zinc	µg/L	22 (18 - 45)	20 (15 - 28)	21 (18 - 34)
Flow	L/s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Pai Min Kok Stream and Kau Wa Keng Stream in 2017

Parameter	Unit	Pai Min Kok Stream		Kau Wa Keng Stream
		AN1	AN2	KW3
Dissolved oxygen	mg/L	8.4 (7.9 - 10.7)	8.7 (8.0 - 10.9)	8.5 (7.6 - 10.4)
pH		7.8 (7.5 - 8.2)	7.9 (7.2 - 8.6)	7.4 (7.2 - 7.9)
Suspended solids	mg/L	2.3 (0.7 - 15.0)	2.2 (1.2 - 35.0)	4.5 (1.4 - 110.0)
5-day Biochemical Oxygen Demand	mg/L	1.1 (0.6 - 7.7)	0.6 (0.2 - 1.4)	2.2 (1.0 - 9.4)
Chemical Oxygen Demand	mg/L	6 (3 - 29)	4 (<2 - 11)	9 (4 - 17)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.9)
<i>E. coli</i>	counts/ 100mL	4 600 (80 - 1 100 000)	3 800 (300 - 270 000)	42 000 (2 000 - 110 000)
Faecal coliforms	counts/ 100mL	22 000 (190 - 1 500 000)	18 000 (3 100 - 450 000)	120 000 (20 000 - 490 000)
Ammonia-nitrogen	mg/L	0.078 (0.011 - 0.630)	0.016 (<0.005 - 0.067)	0.520 (0.190 - 1.200)
Nitrate-nitrogen	mg/L	0.945 (0.320 - 1.800)	0.335 (0.250 - 1.200)	2.100 (0.720 - 3.500)
Total Kjeldahl nitrogen	mg/L	0.46 (0.15 - 1.40)	0.26 (<0.05 - 0.56)	0.91 (0.45 - 2.10)
Ortho-phosphate Phosphorus	mg/L	0.045 (0.007 - 0.870)	0.038 (0.006 - 0.130)	0.061 (0.007 - 0.140)
Total phosphorus	mg/L	0.08 (0.04 - 0.93)	0.06 (0.05 - 0.14)	0.13 (0.04 - 0.43)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	88 (58 - 281)	66 (52 - 248)	115 (<50 - 1 068)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	1.1 (0.5 - 2.0)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	<1 (<1 - 3)
Copper	µg/L	5 (2 - 24)	2 (1 - 12)	5 (2 - 14)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 5)	2 (1 - 24)
Zinc	µg/L	31 (14 - 49)	21 (11 - 48)	104 (62 - 255)
Flow	L/s	NM	6 (3 - 60)	20 (11 - 350)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sam Dip Tam Stream in 2017

Parameter	Unit	Sam Dip Tam Stream		
		TW1	TW2	TW3
Dissolved oxygen	mg/L	8.2 (7.7 - 9.6)	8.6 (8.0 - 10.3)	8.4 (8.0 - 10.2)
pH		7.5 (7.1 - 7.9)	7.8 (7.3 - 8.2)	7.7 (7.2 - 8.2)
Suspended solids	mg/L	1.4 (0.7 - 8.7)	1.3 (0.6 - 3.5)	2.0 (0.5 - 10.0)
5-day Biochemical Oxygen Demand	mg/L	0.7 (0.3 - 1.3)	0.9 (0.4 - 5.2)	0.9 (0.6 - 2.4)
Chemical Oxygen Demand	mg/L	3 (<2 - 5)	5 (<2 - 10)	4 (<2 - 7)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	9 200 (400 - 33 000)	11 000 (2 400 - 43 000)	8 100 (2 000 - 43 000)
Faecal coliforms	counts/ 100mL	47 000 (4 900 - 400 000)	35 000 (8 500 - 85 000)	24 000 (9 400 - 120 000)
Ammonia-nitrogen	mg/L	0.017 (0.008 - 0.140)	0.140 (0.030 - 1.000)	0.037 (0.020 - 0.110)
Nitrate-nitrogen	mg/L	0.760 (0.270 - 0.870)	1.045 (0.370 - 1.700)	1.350 (0.590 - 2.100)
Total Kjeldahl nitrogen	mg/L	0.19 (0.11 - 0.52)	0.30 (0.22 - 1.50)	0.29 (0.14 - 0.61)
Ortho-phosphate Phosphorus	mg/L	0.038 (0.004 - 0.067)	0.094 (0.022 - 0.320)	0.095 (0.022 - 0.180)
Total phosphorus	mg/L	0.06 (0.02 - 0.09)	0.11 (0.03 - 0.37)	0.11 (0.06 - 0.18)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 76)	<50 (<50 - 90)	<50 (<50 - 96)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 3)	2 (<1 - 6)	2 (1 - 6)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 1)
Zinc	µg/L	15 (<10 - 18)	17 (<10 - 22)	18 (11 - 25)
Flow	L/s	NM	58 (32 - 180)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kai Tak River in 2017 (Part 1 of 2)

Parameter	Unit	Kai Tak River		
		KN1	KN2	KN3
Dissolved oxygen	mg/L	6.2 (5.2 - 7.2)	6.9 (6.1 - 7.4)	7.0 (5.8 - 7.4)
pH		7.3 (7.1 - 7.5)	7.4 (7.1 - 7.6)	7.3 (7.3 - 7.5)
Suspended solids	mg/L	6.0 (2.0 - 12.0)	5.9 (2.6 - 28.0)	12.0 (2.6 - 39.0)
5-day Biochemical Oxygen Demand	mg/L	3.3 (1.9 - 4.2)	5.1 (1.7 - 7.1)	5.2 (1.6 - 11.0)
Chemical Oxygen Demand	mg/L	26 (14 - 37)	29 (13 - 34)	24 (16 - 40)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.7)
<i>E. coli</i>	counts/ 100mL	60 000 (11 000 - 310 000)	35 000 (6 000 - 300 000)	39 000 (10 000 - 350 000)
Faecal coliforms	counts/ 100mL	180 000 (34 000 - 1 400 000)	97 000 (38 000 - 550 000)	110 000 (23 000 - 460 000)
Ammonia-nitrogen	mg/L	1.350 (0.670 - 3.500)	1.250 (0.540 - 5.000)	1.400 (0.750 - 5.000)
Nitrate-nitrogen	mg/L	4.000 (2.100 - 5.600)	4.250 (2.800 - 6.500)	3.700 (2.600 - 6.200)
Total Kjeldahl nitrogen	mg/L	2.20 (1.60 - 3.90)	2.25 (1.50 - 5.80)	2.40 (1.80 - 6.40)
Ortho-phosphate Phosphorus	mg/L	1.450 (0.840 - 1.800)	1.700 (1.000 - 2.100)	1.800 (0.710 - 2.100)
Total phosphorus	mg/L	1.70 (0.92 - 1.90)	1.80 (1.30 - 2.30)	1.80 (0.91 - 2.70)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	67 (<50 - 161)	56 (<50 - 292)	61 (53 - 449)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 3)	1 (<1 - 2)	1 (1 - 2)
Copper	µg/L	5 (4 - 16)	6 (4 - 11)	7 (5 - 12)
Lead	µg/L	<1 (<1 - 9)	<1 (<1 - 5)	<1 (<1 - 8)
Zinc	µg/L	31 (20 - 56)	30 (23 - 63)	37 (29 - 92)
Flow	L/s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
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 3. NM indicates no measurement taken.
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Summary of water quality monitoring data for Kai Tak River in 2017 (Part 2 of 2)

Parameter	Unit	Kai Tak River		
		KN4	KN5	KN7
Dissolved oxygen	mg/L	6.6 (5.6 - 7.5)	7.4 (7.0 - 7.9)	7.2 (6.6 - 8.0)
pH		7.3 (7.2 - 7.6)	7.3 (6.9 - 7.4)	7.2 (6.9 - 7.3)
Suspended solids	mg/L	7.8 (2.0 - 50.0)	10.6 (3.6 - 24.0)	4.5 (2.7 - 25.0)
5-day Biochemical Oxygen Demand	mg/L	3.8 (2.3 - 5.4)	5.9 (2.6 - 11.0)	8.6 (2.1 - 19.0)
Chemical Oxygen Demand	mg/L	28 (15 - 34)	31 (14 - 37)	25 (12 - 39)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)
<i>E. coli</i>	counts/ 100mL	19 000 (400 - 180 000)	21 000 (7 000 - 85 000)	15 000 (810 - 250 000)
Faecal coliforms	counts/ 100mL	58 000 (1 900 - 420 000)	61 000 (30 000 - 120 000)	34 000 (3 100 - 540 000)
Ammonia-nitrogen	mg/L	1.000 (0.520 - 4.900)	0.750 (0.270 - 4.800)	0.805 (0.220 - 3.400)
Nitrate-nitrogen	mg/L	3.600 (1.500 - 6.000)	4.800 (2.600 - 5.800)	5.050 (2.300 - 6.600)
Total Kjeldahl nitrogen	mg/L	2.10 (1.40 - 5.00)	2.20 (1.30 - 5.30)	2.25 (1.20 - 5.10)
Ortho-phosphate Phosphorus	mg/L	1.600 (0.370 - 2.000)	1.700 (1.300 - 2.000)	1.700 (1.200 - 2.000)
Total phosphorus	mg/L	1.80 (0.49 - 2.10)	1.75 (1.40 - 2.20)	1.90 (1.40 - 2.50)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.06)
Aluminium	µg/L	58 (<50 - 365)	67 (<50 - 249)	<50 (<50 - 123)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 2)	1 (<1 - 3)	<1 (<1 - 3)
Copper	µg/L	6 (3 - 10)	6 (4 - 18)	7 (4 - 10)
Lead	µg/L	<1 (<1 - 9)	<1 (<1 - 3)	<1 (<1 - 2)
Zinc	µg/L	33 (20 - 85)	32 (21 - 52)	32 (25 - 53)
Flow	L/s	240 (60 - 480)	4 675 (1 890 - 15 050)	260 (114 - 1 110)

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Compliance with the river Water Quality Objectives in 2017

Watercourse	pH	5-Day Biochemical Oxygen Demand	Chemical Oxygen Demand	Dissolved Oxygen	Suspended Solids	Overall Compliance
Eastern New Territories						
Shing Mun River	84%	85%	96%	100%	100%	93%
Lam Tsuen River	100%	81%	94%	99%	100%	95%
Tai Po River	100%	100%	100%	100%	100%	100%
Tai Po Kau Stream	100%	100%	100%	100%	100%	100%
Shan Liu Stream	100%	100%	100%	100%	100%	100%
Tung Tze Stream	100%	100%	100%	100%	100%	100%
Ho Chung River	100%	96%	100%	100%	100%	99%
Sha Kok Mei Stream	100%	88%	100%	100%	100%	98%
Tai Chung Hau Stream	92%	83%	100%	100%	100%	95%
Tseng Lan Shue Stream	100%	53%	97%	100%	100%	90%
Northwestern New Territories						
River Indus	75%	58%	67%	89%	100%	78%
River Beas	100%	44%	81%	100%	100%	85%
River Ganges	100%	67%	89%	100%	100%	91%
Yuen Long Creek	100%	6%	19%	63%	75%	53%
Kam Tin River	100%	0%	29%	71%	100%	60%
Tin Shui Wai Nullah	88%	58%	100%	96%	100%	88%
Fairview Park Nullah	100%	50%	58%	92%	0%	60%
Ha Pak Nai Stream	100%	100%	100%	100%	100%	100%
Tai Shui Hang Stream	100%	100%	100%	100%	100%	100%
Pak Nai Stream	100%	100%	100%	100%	100%	100%
Sheung Pak Nai Stream	100%	91%	100%	100%	100%	98%
Ngau Hom Sha Stream	100%	100%	100%	100%	100%	100%
Tsang Kok Stream	100%	100%	100%	100%	100%	100%
Lantau Island						
Mui Wo River	98%	100%	100%	100%	100%	99%
Tung Chung River	100%	81%	100%	100%	100%	96%
Southwestern New Territories & Kowloon						
Tuen Mun River	100%	67%	90%	96%	100%	91%
Pai Min Kok Stream	100%	96%	100%	100%	100%	99%
Kau Wa Keng Stream	100%	83%	100%	100%	100%	97%
Sam Dip Tam Stream	100%	97%	100%	100%	100%	99%
Kai Tak River	100%	58%	66%	100%	100%	85%
Average Compliance (All monitoring stations)	97%	73%	87%	96%	98%	90%

Water Quality Index (WQI) for inland waters of Hong Kong

The Water Quality Index (WQI) is a numerical value used to denote, based on monitoring data, the general ecological health of a river. It is relevant to conserving the primary beneficial use of rivers for maintenance of aquatic life. The WQI is calculated based on three key parameters: dissolved oxygen, 5-day biochemical oxygen demand (BOD₅) and ammonia-nitrogen content. The measurements of these parameters are evaluated using the following table:

Calculation of Water Quality Index

No. of points awarded	Dissolved oxygen (% saturation)	BOD ₅ (mg/L)	Ammonia-nitrogen (mg/L)
1	91 – 110	< 3	< 0.5
2	71 – 90 111 – 120	3.1 – 6.0	0.5 – 1.0
3	51 – 70 121 – 130	6.1 – 9.0	1.1 – 2.0
4	31 – 50	9.1 – 15.0	2.1 – 5.0
5	< 30 or > 130	> 15.0	> 5.0

All parameters carry equal weighting and the monthly index is the sum of points of the three parameters. The annual WQI of a river station is the average of all monthly WQI in the year. The WQI values range from 3 to 15, corresponding to the following water quality conditions:

Water Quality Index gradings

WQI	Water quality condition
3.0 – 4.5	Excellent
4.6 – 7.5	Good
7.6 – 10.5	Fair
10.6 – 13.5	Bad
13.6 – 15.0	Very Bad

Summary of river water quality improvements in 18 districts

District	Improvement	Remaining Problem
New Territories		
Kwai Tsing	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Water quality in Kau Wa Keng Stream improved from “Bad” in 1991 to “Excellent” in 2017. Its <i>E. coli</i> level has been reduced by over 60% compared with 1998. Sewerage works for villages are being planned.	Still affected by expedient connections and a small number of unsewered villages
Tsuen Wan	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges have been eliminated under the Livestock Waste Control Scheme. Sam Dip Tam Stream improved from “Fair” to “Good” in 1988 to “Excellent” in 2017. Its <i>E. coli</i> level has been reduced by 80% compared with 1999. Pai Min Kok Stream improved from “Bad” in 1988 to “Excellent” in 2017. Its <i>E. coli</i> level has been reduced by over 90% compared with 1989. Sewerage works at hinterland of affected beaches have been completed and progressive connection of unsewered properties to the new public sewers is being made. Sewer provisions for other villages in the area are being planned.	Still affected by expedient connections and a small number of unsewered villages.
Tuen Mun	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. Tuen Mun River upstream improved from “Very Bad” in 1988 to “Bad” to “Good” in 2017. Its <i>E. coli</i> level has been reduced by over 90% compared with 1988. Tuen Mun River mid-stream and downstream improved from “Bad” in 1988 to “Excellent” and “Good”, respectively, in 2017. The <i>E. coli</i> level has been reduced by over 90% compared with 1988. A new trunk sewer to the west of Tuen Mun River was completed in 2015. Sewerage works for villages are progressively being planned and implemented. The Pillar Point Sewage Treatment Works have been upgraded to chemically enhanced primary treatment level with UV disinfection since May 2014.	Still affected by expedient connections and unsewered villages especially in the upstream region.
Yuen Long	Under the Livestock Waste Control Scheme, most of the farms had ceased operation. Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town center are connected to government sewers. Yuen Long Creek and Kam Tin River both changed from “Very Bad” in 1986 to “Bad” to “Fair” in 2017. The <i>E. coli</i> level of these rivers has been reduced by around 70% compared with 1998. Tin Shui Wai Nullah improved from “Bad” to “Fair” in 1993 to “Fair” to “Good” in 2017. The <i>E. coli</i> level of this nullah has been reduced by over 90% compared with 1992. Trunk sewers, pumping stations, and village sewerage are being planned and constructed progressively. The sewage treatment works at San Wai and Yuen Long will be expanded in anticipation of projected population growth, and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms.

District	Improvement	Remaining Problem
New Territories		
North	Under the Livestock Waste Control Scheme, most farms have ceased operation. Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. The downstream stations of River Indus, Beas and Ganges improved from "Very Bad" in 1987 to "Fair" to "Good" in 2017. The <i>E. coli</i> level of these rivers has been reduced by over 90% compared with 1990. Sewerage works for villages are progressively being planned and implemented. The sewage treatment works at Sha Tau Kok and Shek Wu Hui Sewage Treatment Works will be expanded in anticipation of projected population growth and establishment of the New Development Areas and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms.
Tai Po	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of downstream stations of rivers in the area improved from involving "Very Bad" in 1986 to "Good" to "Excellent" in 2017. The <i>E. coli</i> level of the downstream stations of Lam Tsuen River and Tai Po River has been reduced by over 60% and 85%, respectively, compared with 1999. Sewerage works for villages are progressively being planned and implemented. The Tai Po Sewage Treatment Works acquired UV disinfection facilities in 2010 and its treatment capacity has been increased to 120 000 m ³ /d since 2015.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sha Tin	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town center are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of the Shing Mun River main channel improved from "Fair" in 1986 to "Excellent" in 2017. The <i>E. coli</i> level of the main channel has been reduced by over 90% compared with 1988. Village sewerage is being extended to most of the remaining villages in the area. The Sha Tin Sewage Treatment Works acquired UV disinfection facilities in 2010.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sai Kung	Factory discharges are now controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. The rural village houses are being connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. The water quality of the rivers has improved from largely "Bad" in 1991 to "Good" to "Excellent" in 2017. The <i>E. coli</i> level of rivers in Port Shelter WCZ has been reduced by over 60% compared with 1997. Local town and village sewerage improvements are progressively being planned and implemented. Upgrading of the Sai Kung Sewage Treatment Works is under planning.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.

District	Improvement	Remaining Problem
New Territories		
Islands	Some of the domestic premises are connected to government sewers. Under the Livestock Waste Control Scheme, the small number of livestock farms has all been closed. Mui Wo River and Tung Chung River have stayed "Good" or better throughout recent decades. The <i>E. coli</i> level of Mui Wo River has been reduced by over 90% compared with 1988. Additional village sewerage and sewage treatment facilities and upgrading of treatment facilities are being planned and provided to Southern Lantau, Tung Chung West, Mui Wo, Lamma Island, Cheung Chau and Peng Chau.	Still affected by discharges from a small number of unsewered villages and some expedient connections.
Kowloon		
Yau Tsim Mong	The urban area of Kowloon contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified. The water quality of the Kai Tak River improved from mainly "Very Bad" in 1986 to all "Good" in 2017. The <i>E. coli</i> level of the region has been reduced by over 85% compared with 1999. The Upgrading of Central and East Kowloon Sewerage was completed in 2016 to increase the sewerage capacity of the Kai Tak Development Area.	Mainly affected by rear lane washing activities, urban run-offs and expedient connections.
Sham Shui Po		
Kowloon City		
Wong Tai Sin		
Kwun Tong		
Hong Kong Island		
Central and Western	The urban area of Hong Kong Island contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified.	Mainly affected by rear lane washing activities, urban run-offs and expedient connections. Some streams are affected by discharges from squatters built near the streams.
Wan Chai		
Eastern		
Southern		